

BlackRock industry project • UC Berkeley Haas MFE

# Macroeconomic Surprises, DTW Price Patterns and the Intraday Response of ES, NQ and ZN Futures

E-mini S&P 500 (ES) • E-mini Nasdaq-100 (NQ) • 10-Year Treasury Note (ZN)

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# Structure of the presentation

1

## The answer and the questions

Headline results · motivation · 7 research questions

slides 3–4

2

## Markets, signals and method

ES, NQ, ZN · jump vs drift · surprise, DTW, state · walk-forward design

slides 5–9

3

## RQ1–RQ2 · What moves, what is tradeable

Notebooks 02, 02b: CPI dominates; the drift after the first minute

slides 10–12

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## RQ3 · Pre-event market state

Notebooks 03, 03b: volatility and the CPI response

slides 13–14

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## RQ4 · CPI walk-forward strategy

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## RQ5 · Scaling to ~10 releases

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## Discussion and conclusions

Why equities not bonds · final comparison · recommendations · future work

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**Slide 38 · Acknowledgements and questions**

# The answer in one slide

Walk-forward 2019–2026, entry one minute after the release, net of 2 ticks + \$4.50 per round trip

**$\approx 1.0$**

Net Sharpe of the CPI surprise strategy on ES (0.8–1.0 on NQ)

**0.72**

Net Sharpe of the ES + NQ multi-release book, ~195 trades a year

**30 min**

How long the tradeable drift lasts after the first minute

**0 / 0**

Out-of-sample gain from pre-event state / from DTW patterns

## Works

- Macro surprise in equity-index futures (ES, NQ)
- CPI is the core release
- ~100 trades a year per market with ~10 releases

## Does not help

- Pre-event state: predicts the size of the move, not its direction
- DTW price patterns: no directional information
- More windows per release

## Does not work

ZN (10-year T-note): priced in the first minute; every strategy loses after costs

# Why macro releases, and what we asked

*Scheduled information shocks: known time, published expectation, immediate repricing*

## The opportunity and the problem

- The release time and the consensus are known in advance
- Theory predicts the direction: hot inflation → higher rates → lower equities and bond prices
- But the first move happens in milliseconds: a strategy that decides after seeing the number can only trade what comes after the first minute

**RQ1**

Which releases move ES, NQ and ZN, and by how much?

**RQ2**

How much of the reaction is tradeable after the first minute?

**RQ3**

Does the pre-event state change the reaction?

**RQ4**

Can it be traded in real time after costs? Does state help?

**RQ5**

How many trades a year across many releases?

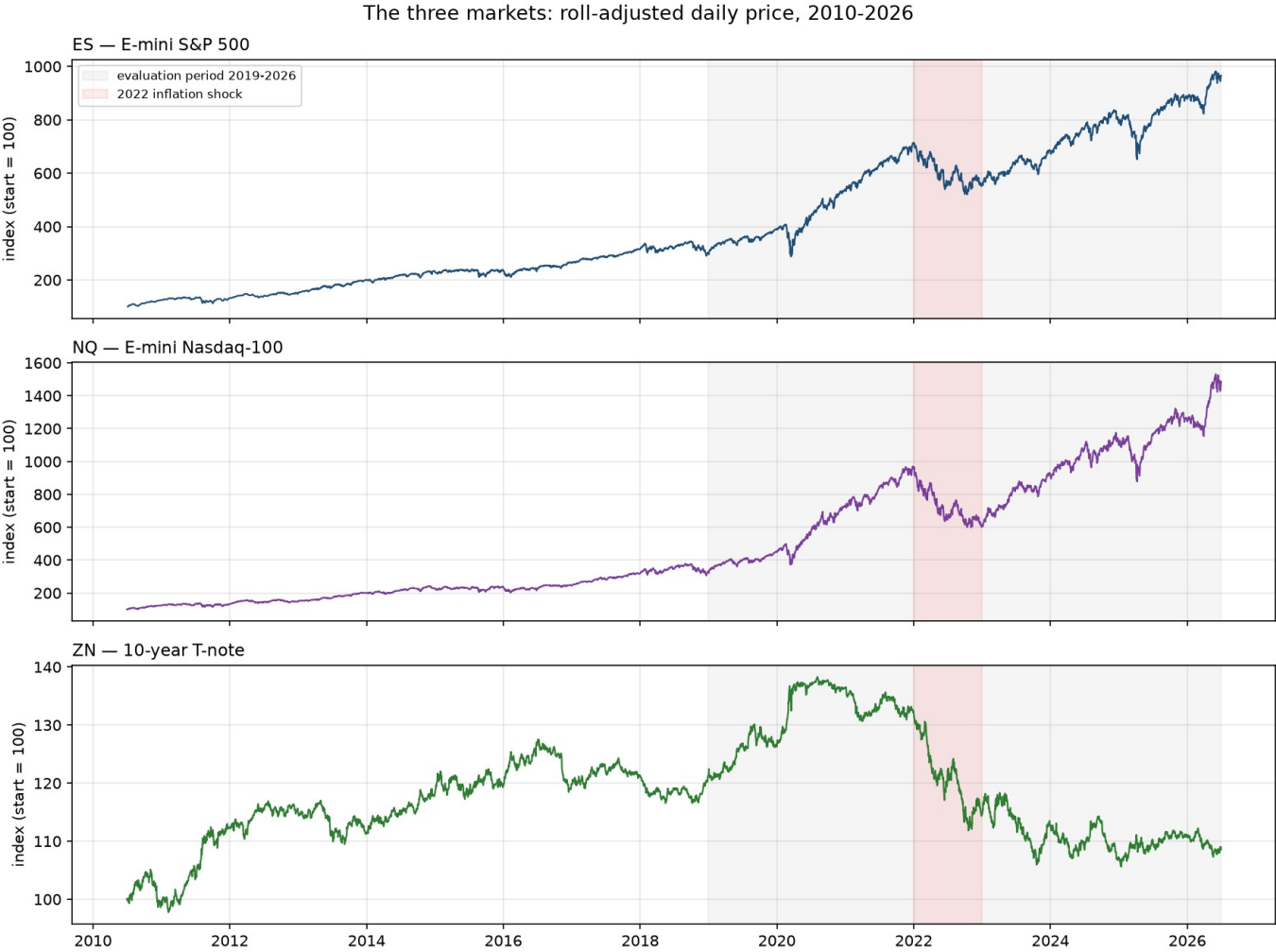
**RQ6**

Does DTW add information? Do more windows help?

**RQ7**

Do markets combine? How does the team compare at equal costs?

# Three markets, three ways to react to the same news



|                         | ES       | NQ         | ZN               |
|-------------------------|----------|------------|------------------|
| Underlying              | S&P 500  | Nasdaq-100 | 10-yr T-note     |
| Value of a tick         | \$12.50  | \$5.00     | \$15.63          |
| Cost (2 ticks + \$4.50) | ≈1.7 bps | ≈0.7 bps   | ≈3.0 bps         |
| Buy & hold 2019–26      | +218%    | +348%      | –10.5%           |
| Hot CPI print           | down     | down more  | down (yields up) |

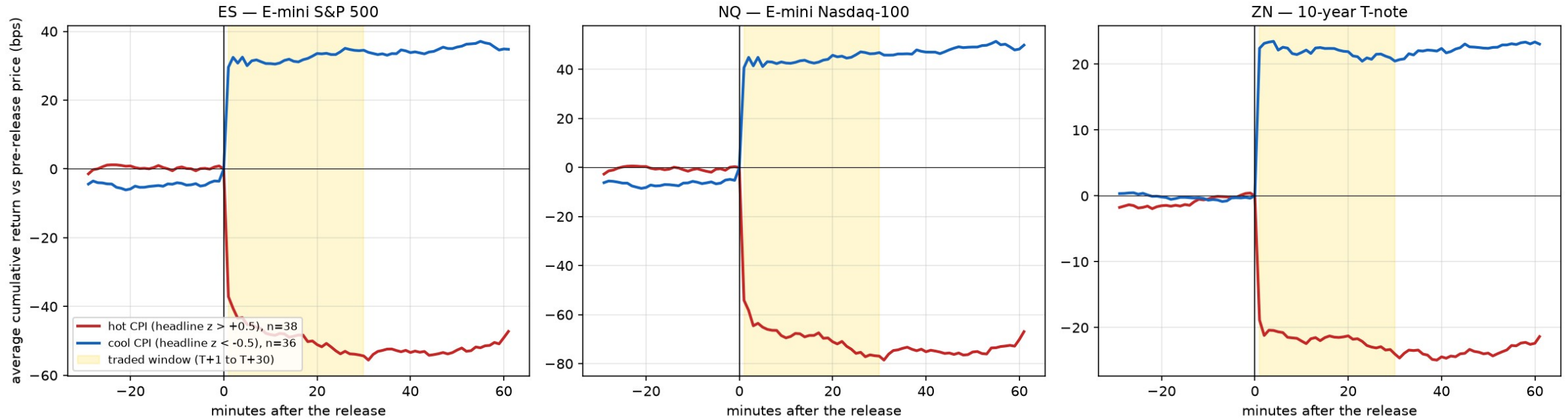
## Why these three

- Is the effect a property of one contract, of equity indices, or of rate-sensitive markets?
- NQ: long-duration tech → more rate-sensitive
- ZN: prices policy expectations directly

# A release produces a jump, then a drift

Average path around 74 CPI releases with  $|\text{surprise}| > 0.5 \text{ SD}$ , 2016–2026

Average market path around CPI releases (2016-2026): a jump at the release, then a smaller drift



**-37 → -55 bps**

ES after a hot CPI print: first-minute jump, then T+30

**-55 → -77 bps**

NQ: larger jump and larger drift

**-19 → -24 bps**

ZN: almost nothing left after the first minute

# Two signal families, plus the market state

## 1 • Macro surprise

- Surprise = actual – Bloomberg consensus
- Standardised causally by the SD of previous surprises ( $\geq 24$ ), capped at  $\pm 5\sigma$
- Line signs from the historical jump; lines released together form a family
- $x > 0$  always means good news for that market (NFP sign reversed for ZN)

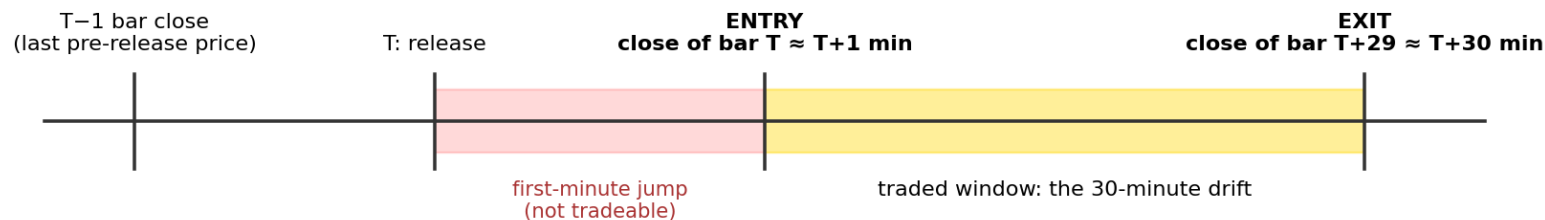
## 2 • DTW price pattern (team spec)

- Query: z-scored 31-minute path ending at entry
- Library: earlier releases at the same clock time, 3-year lookback,  $\leq 600$
- $k = 15$  nearest, band 5, weights  $1/\text{distance} \times 2\text{-year half-life}$
- Trade the neighbours' average next-30-min return if outside the middle 40%

## 3 • Pre-event state

- Six pre-registered states: 20-day vol, 1-day vol, 20-day momentum, 60-day drawdown, overnight move, previous surprise
- Causal percentiles against a trailing one-year window
- Used to scale or filter the surprise trade

# Method: executable, walk-forward, cost-aware



## Entry

Close of the first post-release bar ( $\approx$  T+1 min); exit at T+30. The jump is never traded

## Walk-forward

Every sign, screen, model and gate uses only earlier releases; evaluation 2019–2026

## Costs

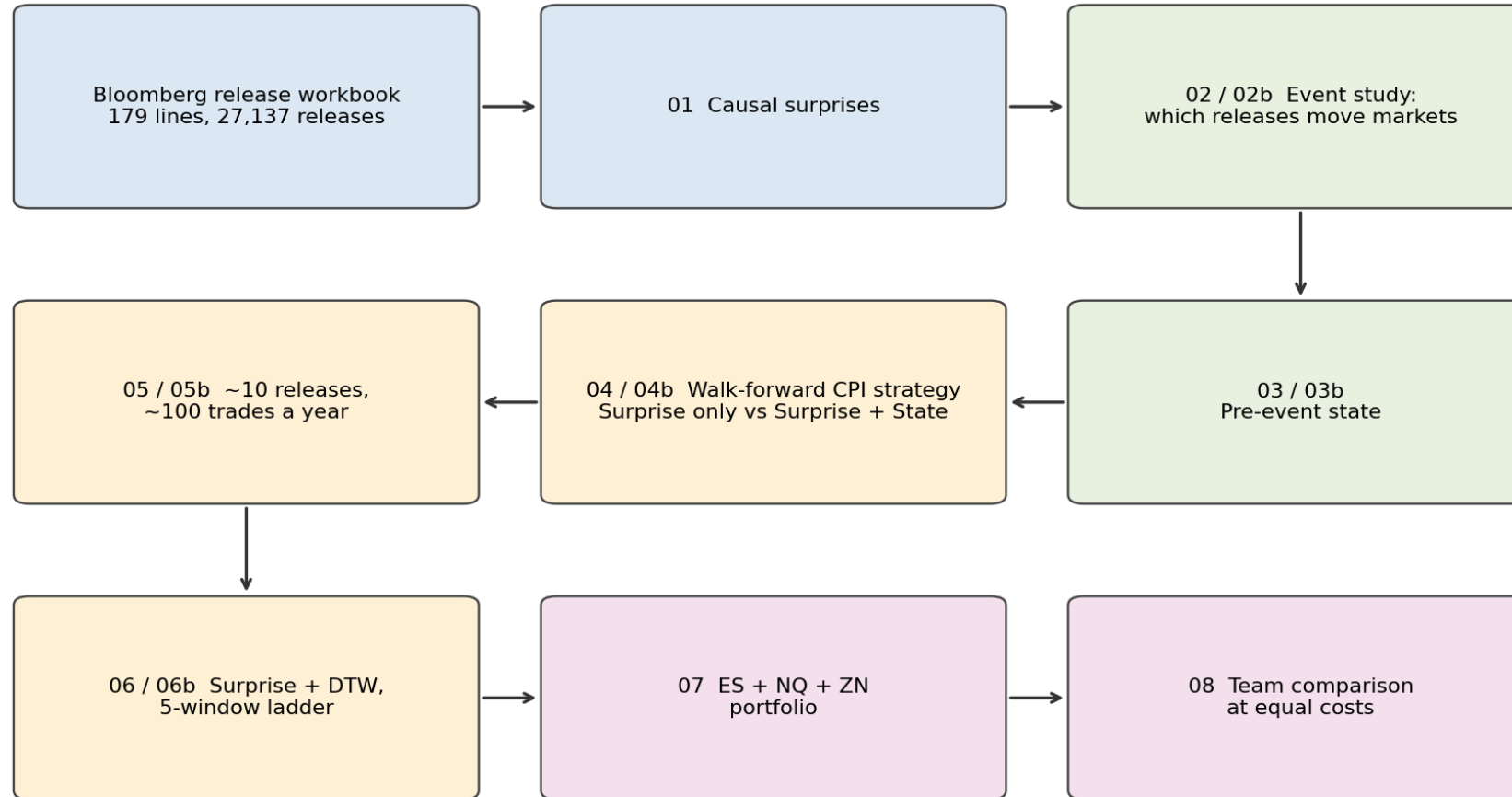
2 ticks + \$4.50 per round trip; sensitivity 0–8 ticks and the team's  $\frac{1}{4}$  tick per side

## Statistics

HC3 errors, BH-FDR, permutation and random-sign tests, block-bootstrap Sharpe

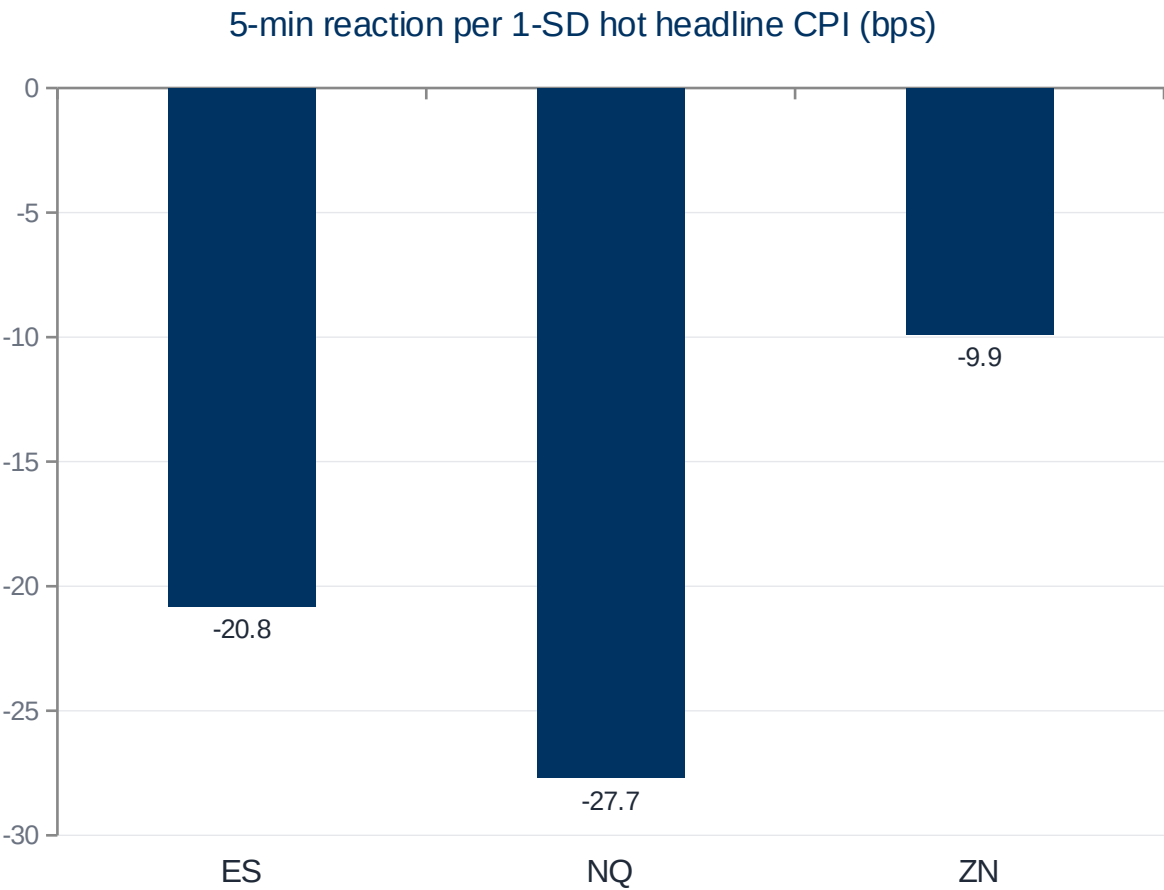
# Research pipeline: 8 notebooks, repeated for NQ and ZN

Market data: ES / NQ / ZN 1-minute bars → roll-adjusted log prices (used from Notebook 02 onwards)



# RQ1 · CPI is the release that moves all three markets

*Notebooks 02 and 02b: 363 CPI, NFP and PCE releases, 2016–2026*



|                            | CPI      | NFP      | PCE      |
|----------------------------|----------|----------|----------|
| Mean   30-min move  , ES   | 38.5 bps | 29.6 bps | 14.7 bps |
| R <sup>2</sup> , 5 min, ES | 0.27     | 0.03     | 0.10     |
| R <sup>2</sup> , 5 min, NQ | 0.30     | 0.01     | 0.10     |
| R <sup>2</sup> , 5 min, ZN | 0.30     | 0.03     | 0.04     |

## Takeaway

- CPI: large move and a predictable direction in every market
- NFP moves markets as much, but its surprise explains only 1–3%: jobs, unemployment and wages often disagree
- NQ reacts ~35% more than ES (rate-sensitive tech)

# RQ1 · How much ES moves, and how much the surprise explains

Notebook 02, ES: 122 CPI, 122 NFP and 121 PCE releases, 2016–2026

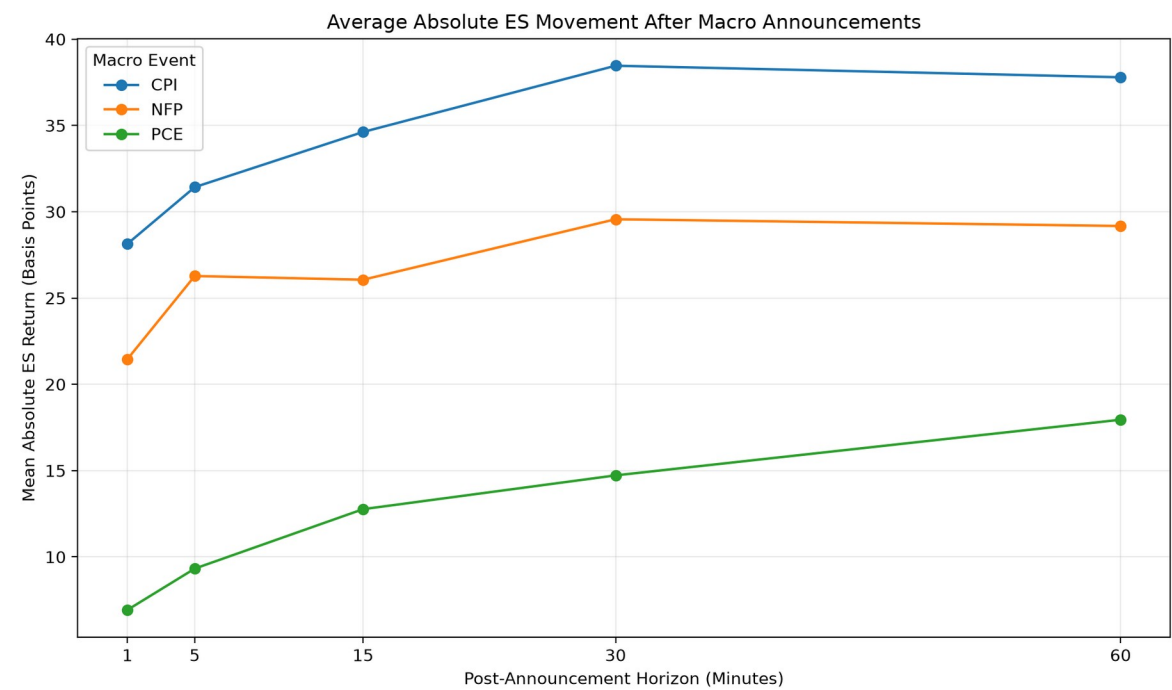


Figure 3 · Mean absolute ES reaction by release and horizon (bps)

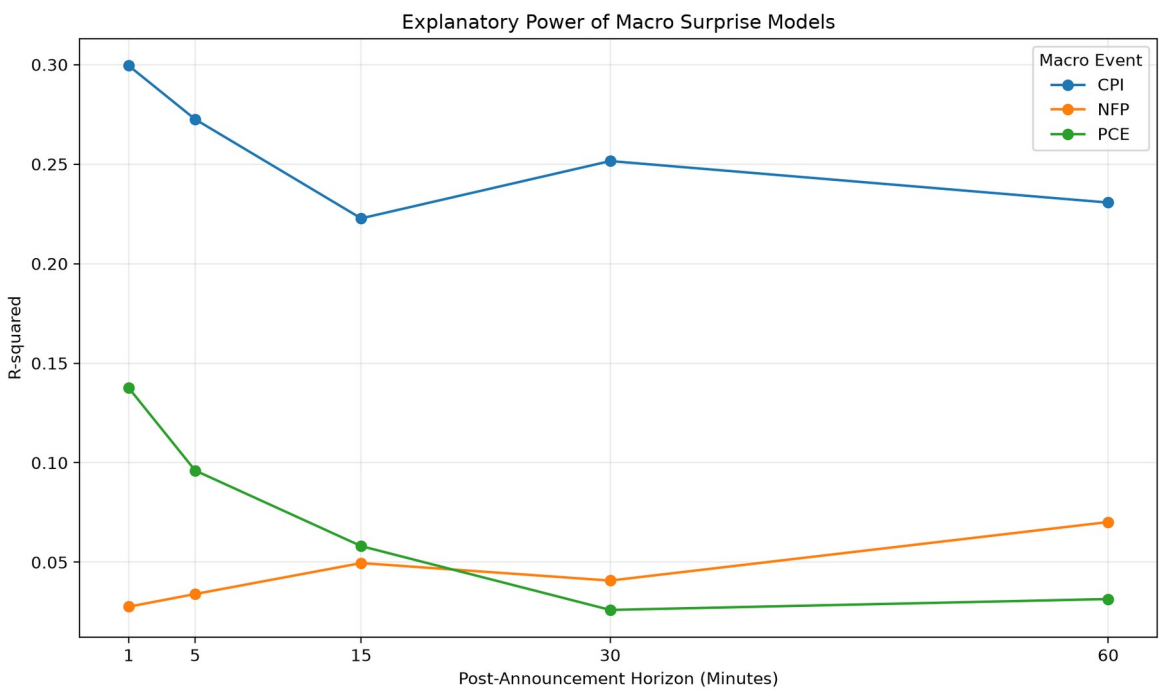
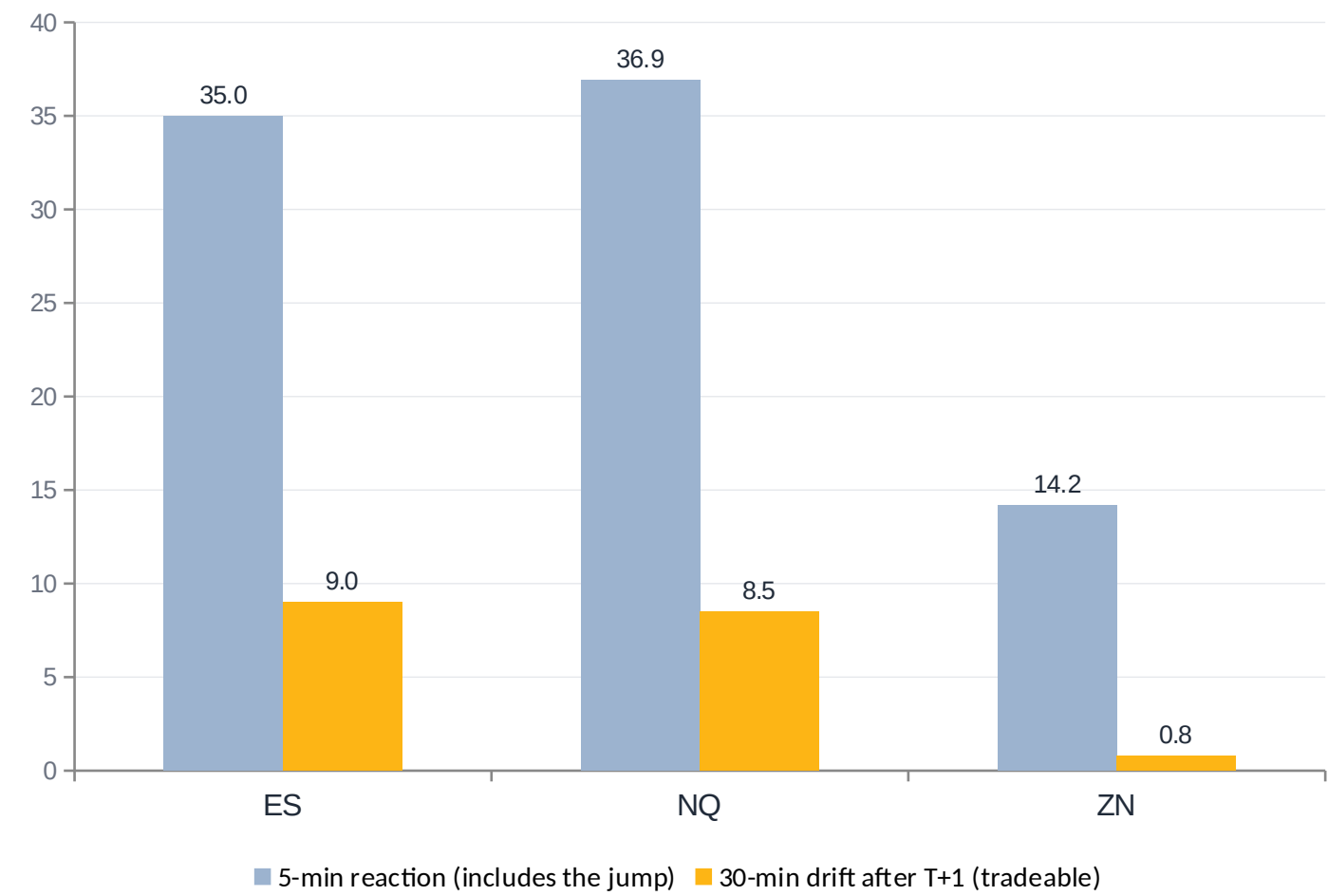


Figure 4 · Share of the ES reaction explained by the surprise ( $R^2$ )

CPI moves ES most (38 bps at 30 min) and its surprise explains 22–30% of the move at every horizon. NFP moves ES almost as much (30 bps) but explains only 3–7%; PCE is informative only in the first minutes.

# RQ2 · Most of the reaction is untradeable

CPI surprise beta (bps per 1 SD): 5-minute reaction from the pre-release price vs 30-minute drift after entry



~1/4

of the CPI reaction remains as a drift in ES and NQ

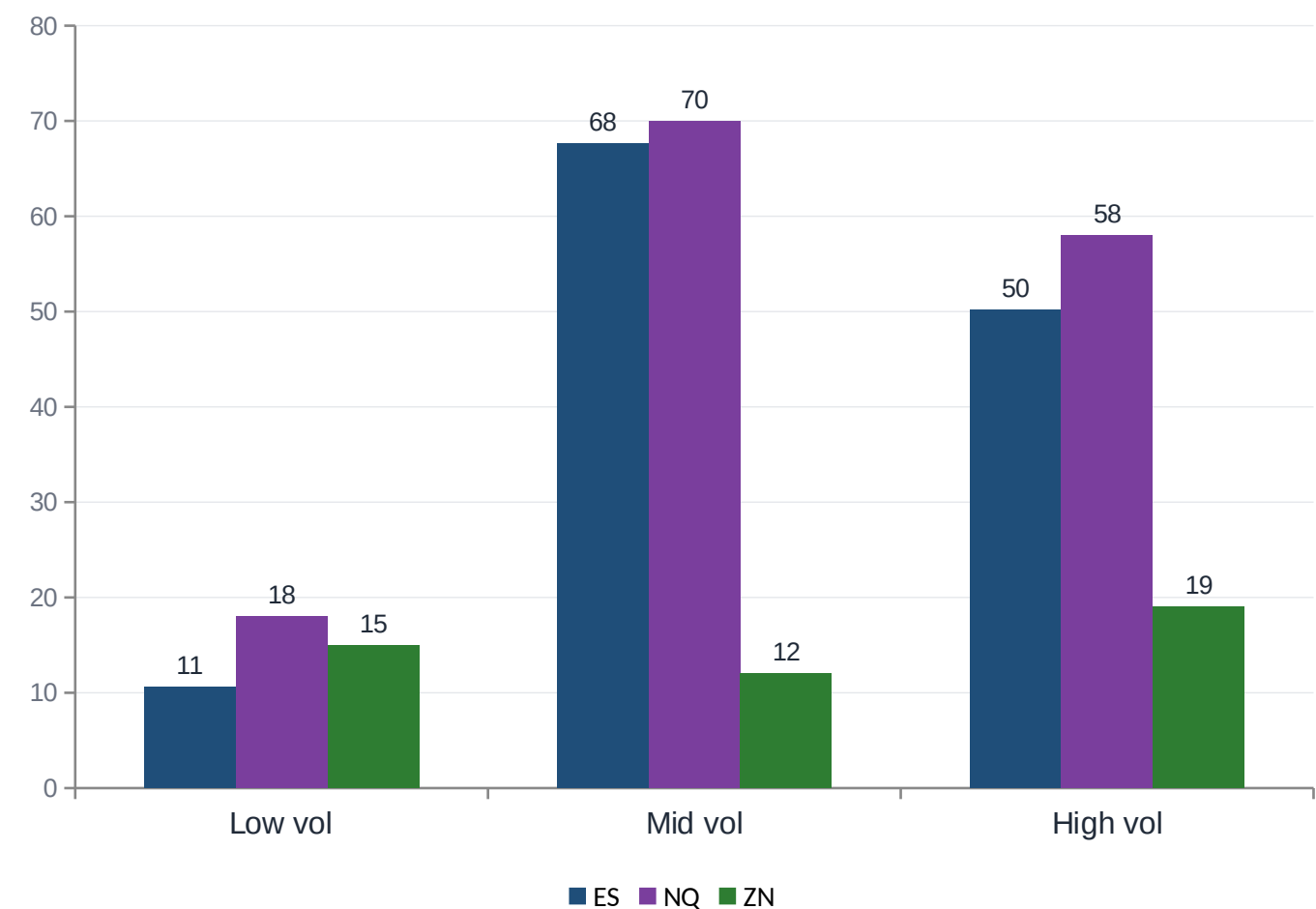
0.8 bps

ZN drift per 1 SD, below its 3 bps cost

Drift vs cost decides the rest of the study: ES 9 vs 1.7 bps, NQ 8.5 vs 0.7 bps, ZN 0.8 vs 3.0 bps.

# RQ3 · In volatile markets CPI hits equities much harder

Notebook 03: CPI surprise beta on the 5-min reaction by tercile of 20-day volatility (bps per 1 SD)

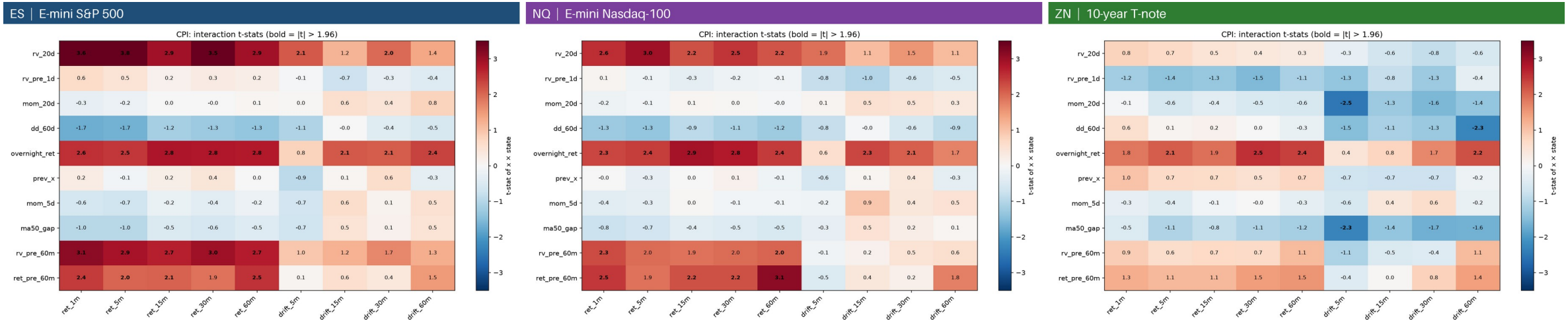


|                          | ES      | NQ      | ZN    |
|--------------------------|---------|---------|-------|
| Best interaction t       | 3.8     | 3.0     | -3.2  |
| Survive FDR              | 2       | 0       | 1     |
| Size tests surviving FDR | 198/360 | 154/360 | 0/360 |

## But

- The effect is mostly on the first-minute jump
- Significant only after 2021
- The state reliably predicts the size of the move, not its direction

# RQ3 · The same pattern in ES and NQ, a different one in ZN

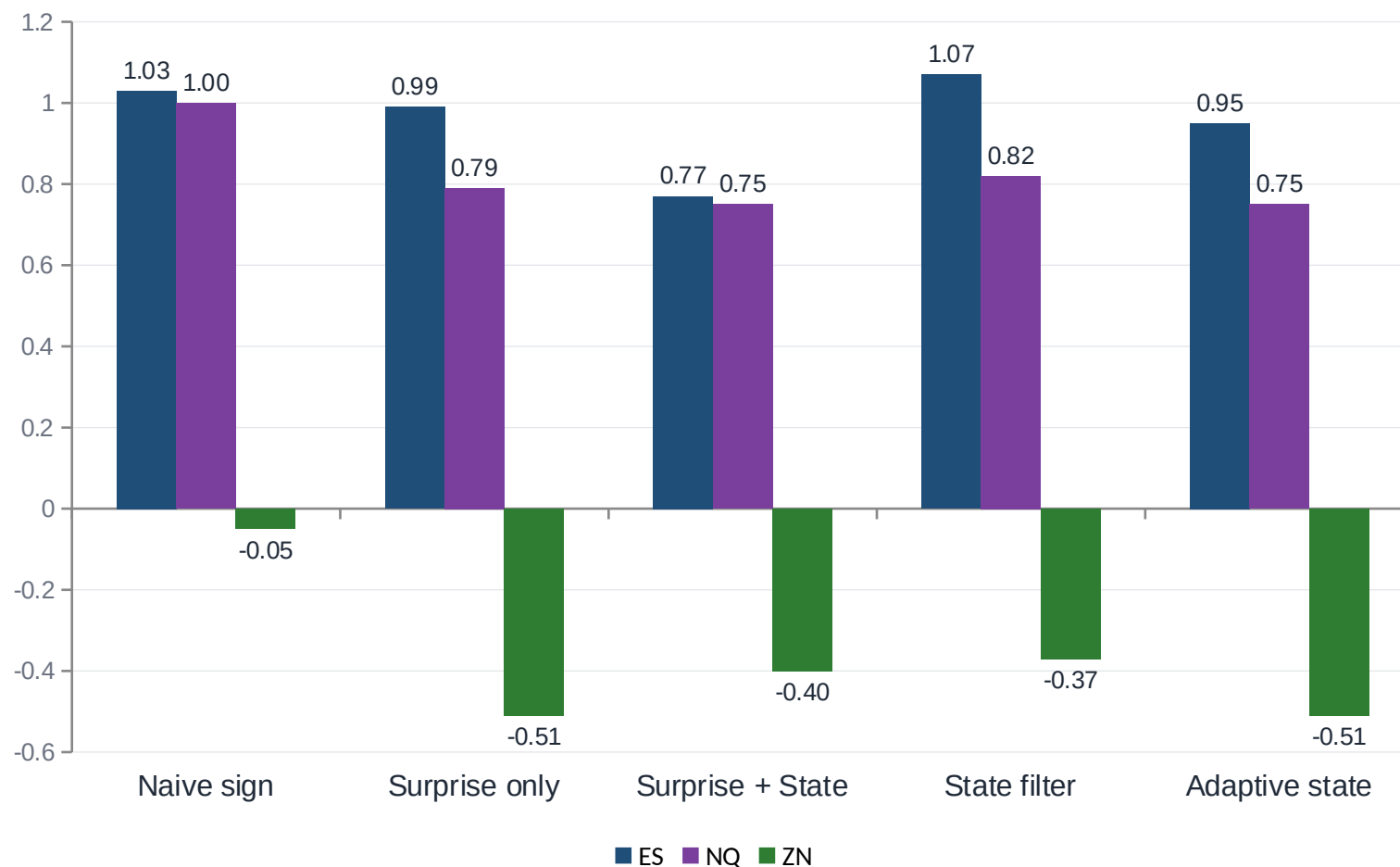


## Reading the t-statistics (CPI × state; rows = states, columns = horizons)

- ES: 20-day volatility row dark red on the reactions (t up to 3.8), much lighter on the drifts
- NQ: the same red rows, slightly weaker (t up to 3.0) — not a fluke of one contract
- Overnight return strengthens the response in all three markets
- ZN: no volatility effect; strong recent trends weaken the post-release drift
- Selected for trading: ES  $rv_{20d} \times CPI$ ; NQ none; ZN 1-day vol × all releases

# RQ4 · CPI surprises are tradeable in ES and NQ, not in ZN

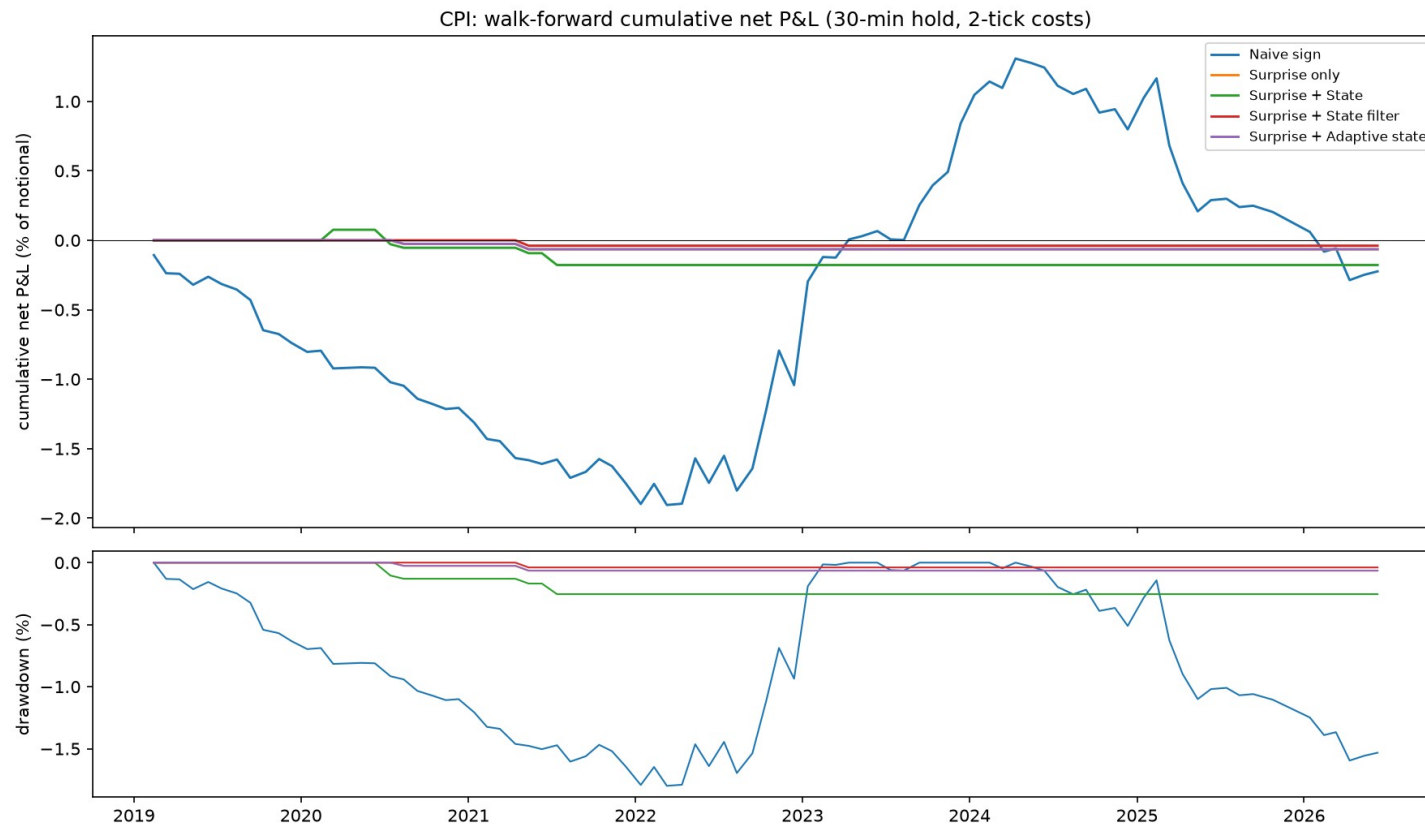
Notebook 04: CPI walk-forward net Sharpe, 2019–2026, 2 ticks + \$4.50



## Takeaways

- ES surprise only: 12.8 bps per trade,  $t = 2.8$ , CI 0.25–1.72
- The naive sign is as good as any fitted model (NQ: significantly better)
- No state model beats surprise only
- ZN: +2.7 bps gross per trade, eaten by the 3 bps cost

# RQ4 · ZN: the fitted models correctly refuse to trade



**+2.7 bps**

Naive CPI gross profit per trade in ZN (gross Sharpe 0.53)

**-0.3 bps**

the same after the 3 bps round-trip cost (Sharpe -0.05)

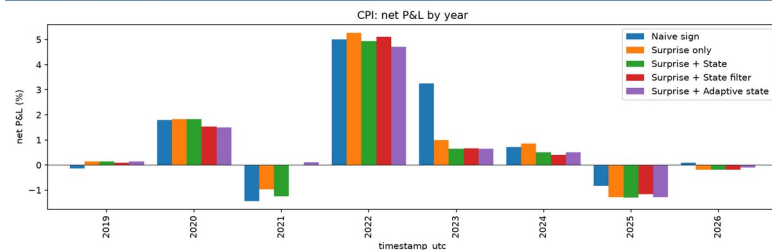
## Why the lines are flat

- The surprise model trades only if the predicted move exceeds the cost
- In ZN that happened 1–5 times in 7.5 years

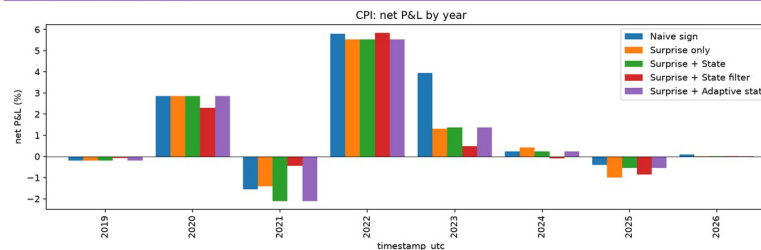
Figure 16 · ZN: walk-forward cumulative net P&L of the CPI strategies and drawdowns

# RQ4 · The edge is concentrated in the 2022 inflation shock

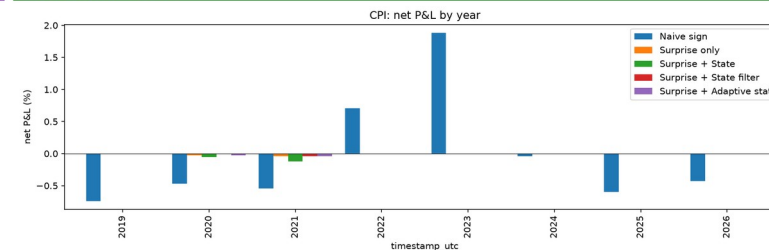
ES E-mini S&P 500



NQ E-mini Nasdaq-100



ZN 10-year T-note



CPI net P&L by year (% of notional), five strategies: ES (left), NQ (centre), ZN (right)

## 79% / 74%

Share of CPI P&L from 2022 (ES / NQ)

## 0.58 / 0.68

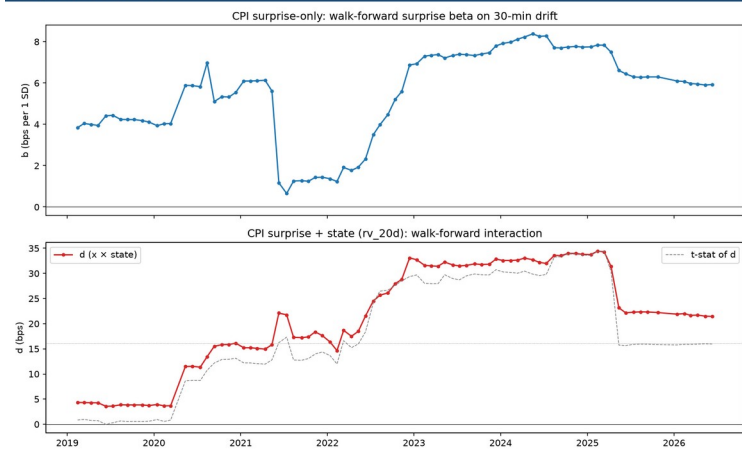
Naive CPI Sharpe excluding 2022 (ES / NQ)

### Why it matters

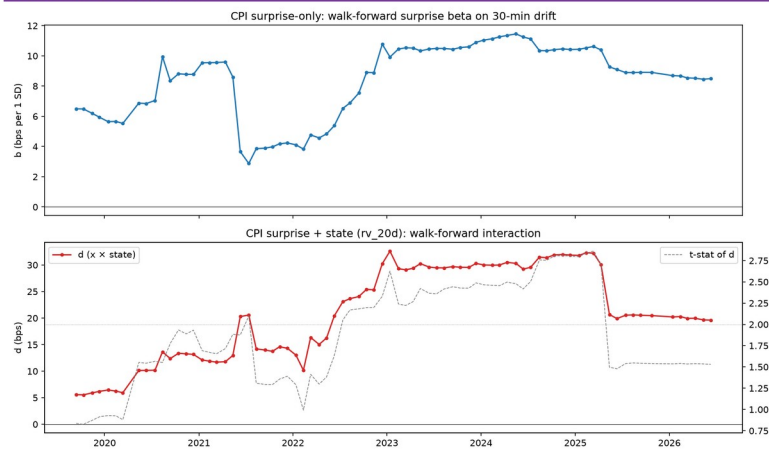
- CPI pays when inflation drives policy expectations
- The state effect was invisible in real time until 2022 ( $t = 0.5$  in 2019, 3.3 in 2022)
- The 'best' state changed every few years: none  $\rightarrow$  20-day vol  $\rightarrow$  overnight move
- Implication: scale exposure with the macro regime, not with a state model

# RQ4 · When could a trader have known about the state effect?

ES | E-mini S&P 500



NQ | E-mini Nasdaq-100



ZN | 10-year T-note

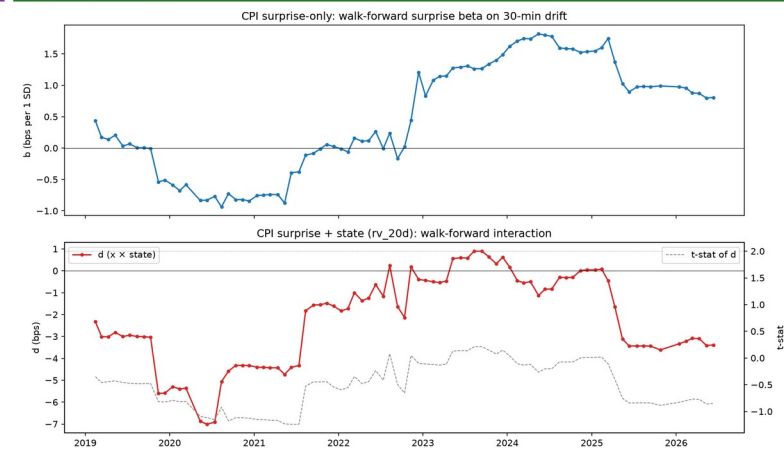


Figure 17 · Walk-forward CPI surprise beta (top) and CPI × 20-day-vol interaction with its t-statistic (bottom): ES, NQ, ZN

## ES

- Beta 4 bps (2019) → ~1 bp (mid-2021) → ~8 bps (2023–24)
- Interaction t crosses 2 only in mid-2022, ~3.5 in 2023–25

## NQ

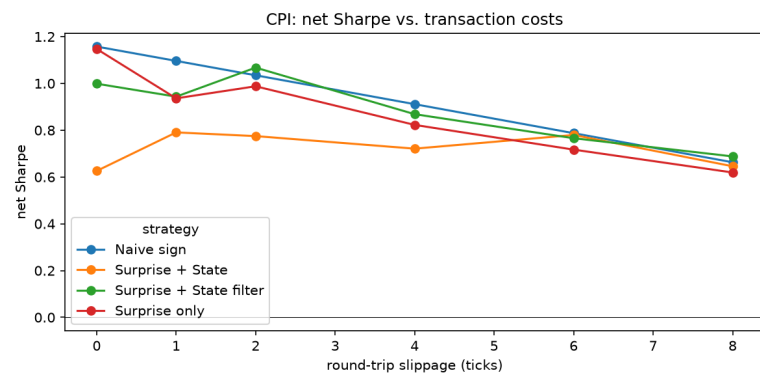
- Same shape at a higher level: 6 → 3 → 10–11 bps
- Interaction t peaks ~2.8 in 2024–25, falls to 1.5

## ZN

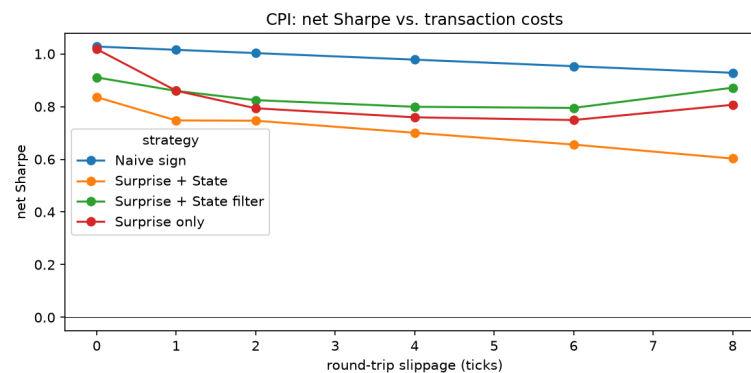
- Beta between –0.9 and +1.8 bps for the whole period
- Interaction never significant (t between –1.2 and +0.2)

# RQ4 · Robust to costs and delays in equities

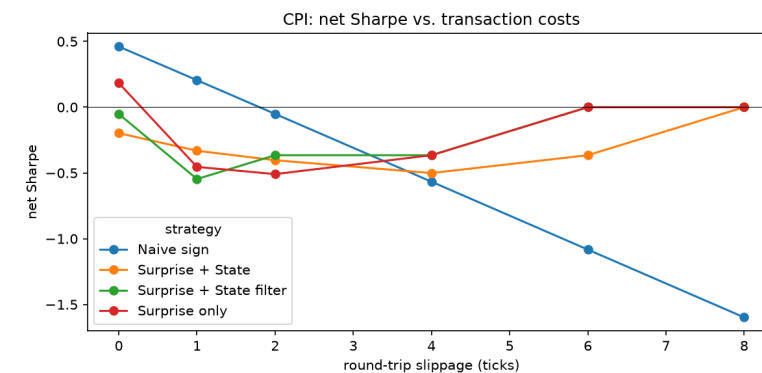
ES | E-mini S&P 500



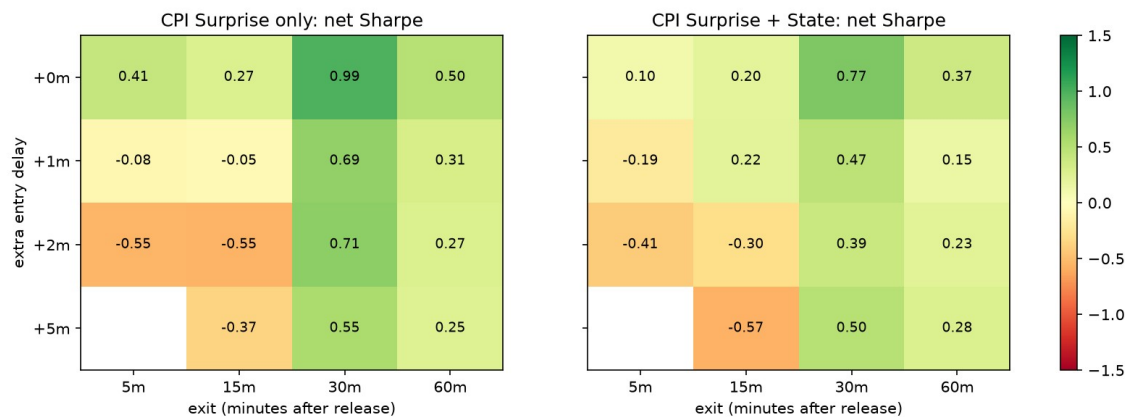
NQ | E-mini Nasdaq-100



ZN | 10-year T-note



CPI net Sharpe vs round-trip slippage (0–8 ticks): ES (left), NQ (centre), ZN (right)



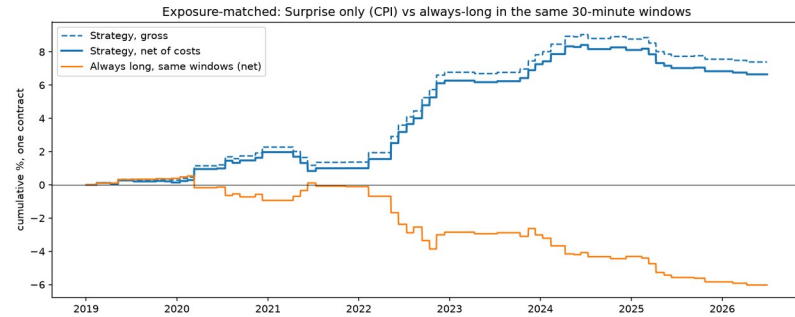
ES: Sharpe by entry delay (rows) and holding period (columns)

## Robustness

- ES: 1.15 → 0.62 from 0 to 8 ticks
- NQ: almost flat (0.75–1.02)
- ZN: turns negative at ~2 ticks
- 30-min hold survives 1–5 min delays; 5-min holds collapse
- A slow 30-minute continuation, not a speed race

# RQ4 · The direction is what earns the money

ES | E-mini S&P 500



**+6.6% vs -6.0%**

ES: CPI surprise strategy vs always long in the same 30-minute windows

**+7.5% vs -5.4%**

NQ: same comparison

NQ | E-mini Nasdaq-100

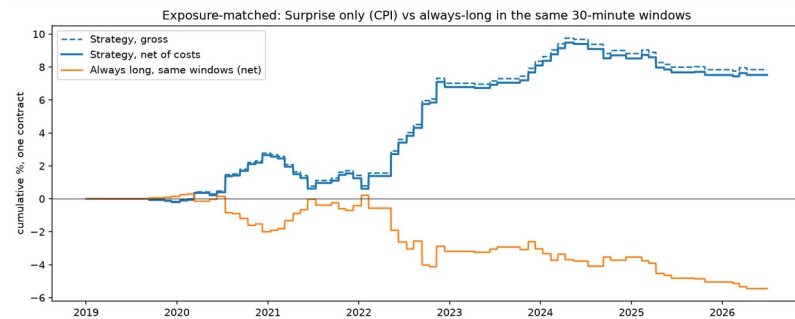
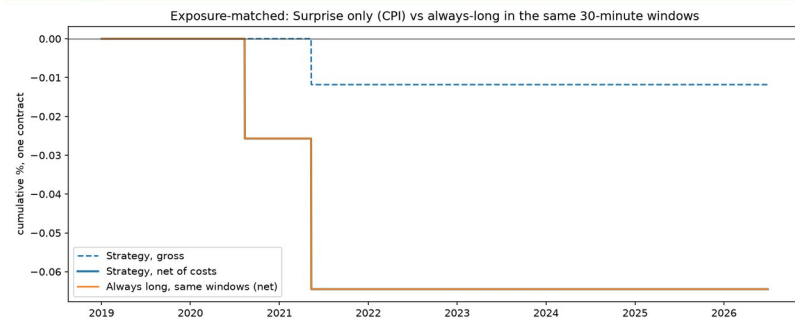


Figure 22 · the peer (DTW team) chart format

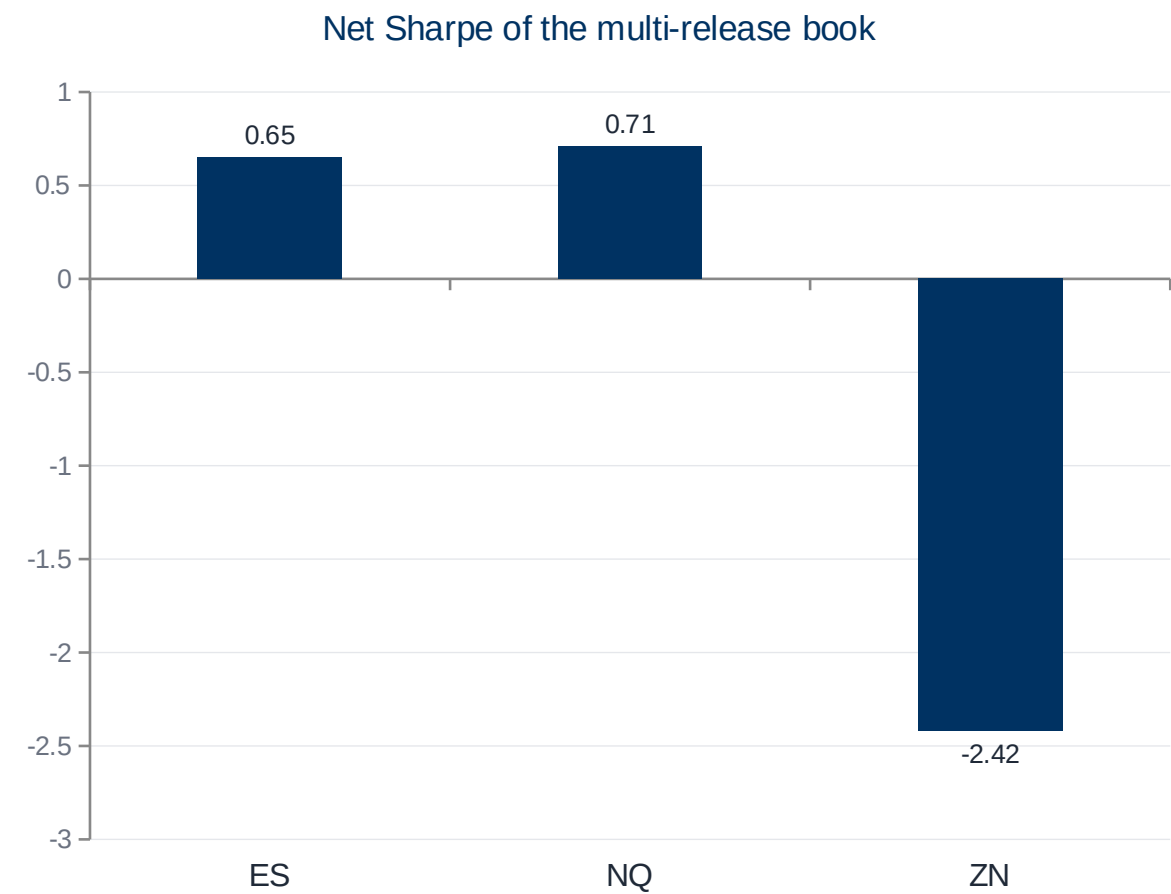
- Solid blue: strategy net of costs; dashed: gross; orange: always long in the same windows
- Being in the market at CPI time earns nothing by itself; the profit comes from the sign of the surprise ( $\approx 13$  percentage points in ES and NQ)
- Gross and net lines are close: costs are not the binding constraint for CPI
- Beta to buy-and-hold  $\approx 0$ : the strategy is market-neutral (0.06–0.1% time in market)

ZN | 10-year T-note



# RQ5 · Ten releases give ~100 trades a year per market

Notebook 05: top-10 families by an impact screen (history only), naive sign, one 30-min trade per release



| All · naive       | ES     | NQ    | ZN     |
|-------------------|--------|-------|--------|
| Trades / year     | 98     | 97    | 82     |
| Net bps / trade   | 1.7    | 2.4   | -3.4   |
| Total net P&L     | 12.8%  | 17.5% | -20.5% |
| Sharpe ex-2022    | 0.28   | 0.44  | -2.53  |
| Random-sign p     | <0.001 | 0.005 | 0.81   |
| Sharpe at 4 ticks | 0.20   | 0.60  | -4.24  |
| 2nd window alone  | -0.79  | -0.23 | -3.08  |

Tenfold more trades with the edge intact in equities; the edge is gone within 30 minutes.

# RQ5 · The multi-release books over time

Figure 23 · ES

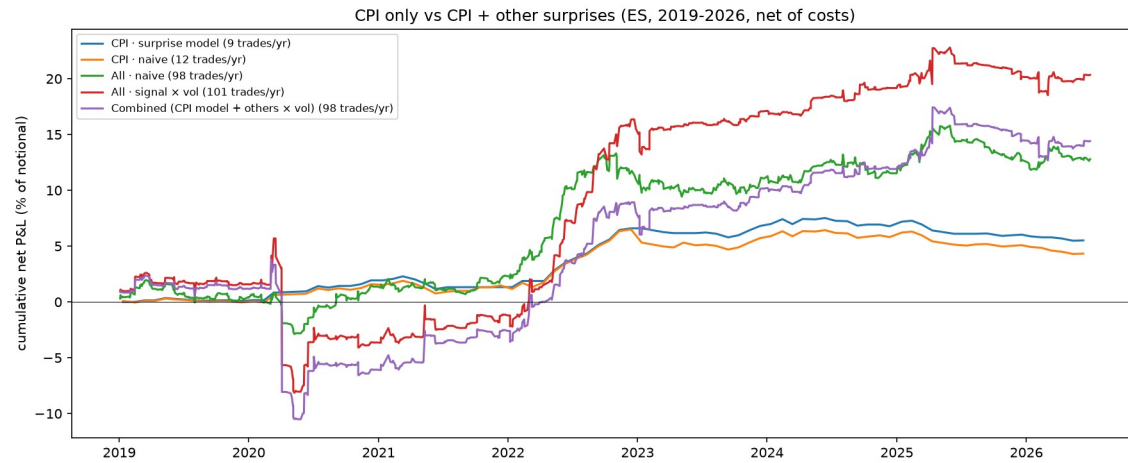


Figure 24 · NQ

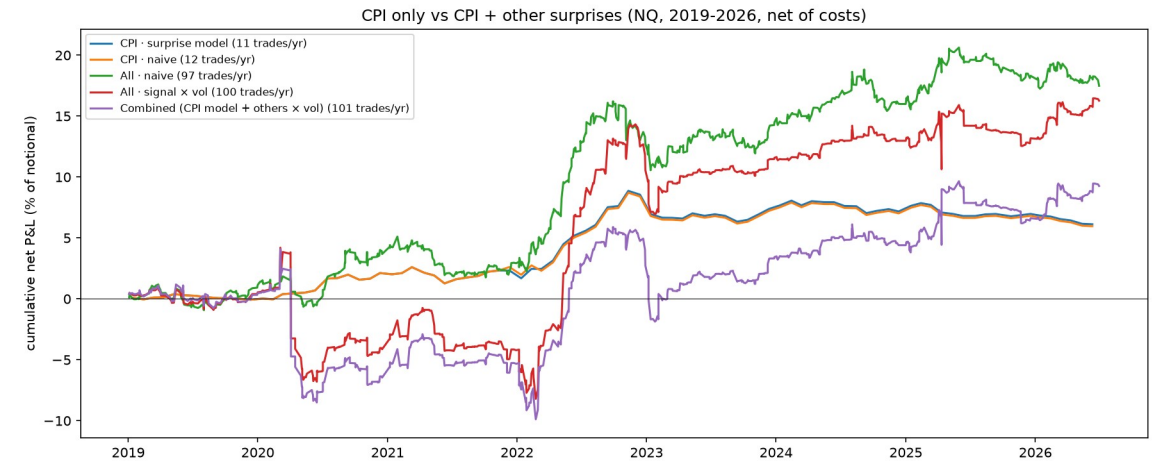
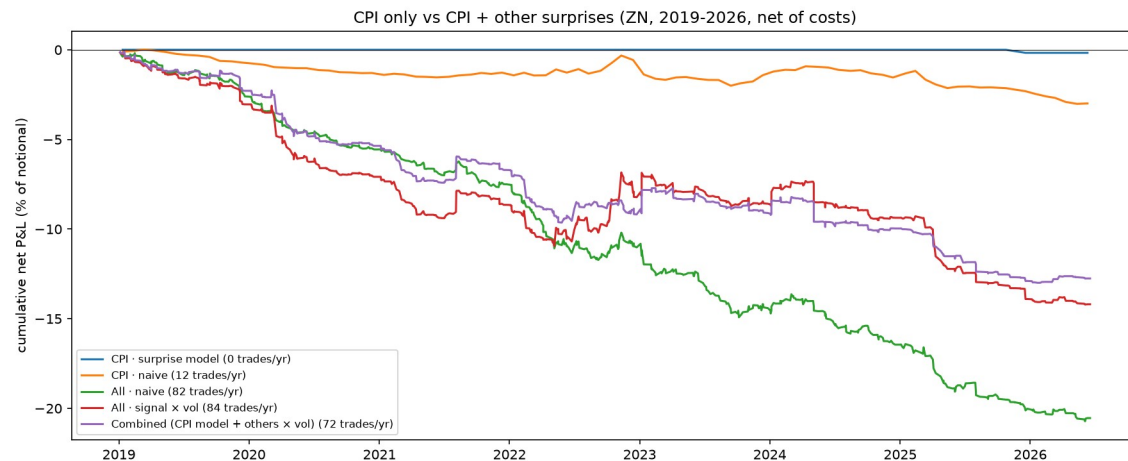


Figure 25 · ZN

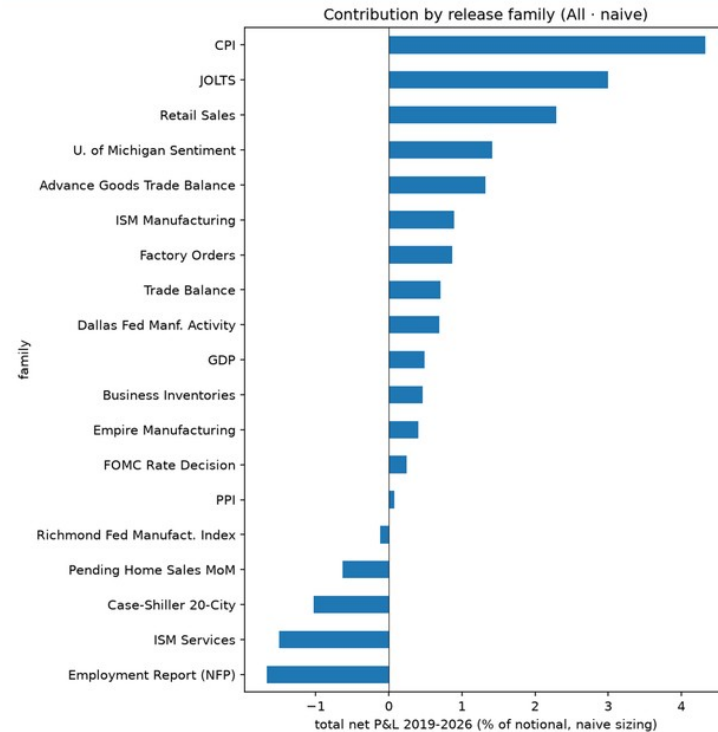


## Reading

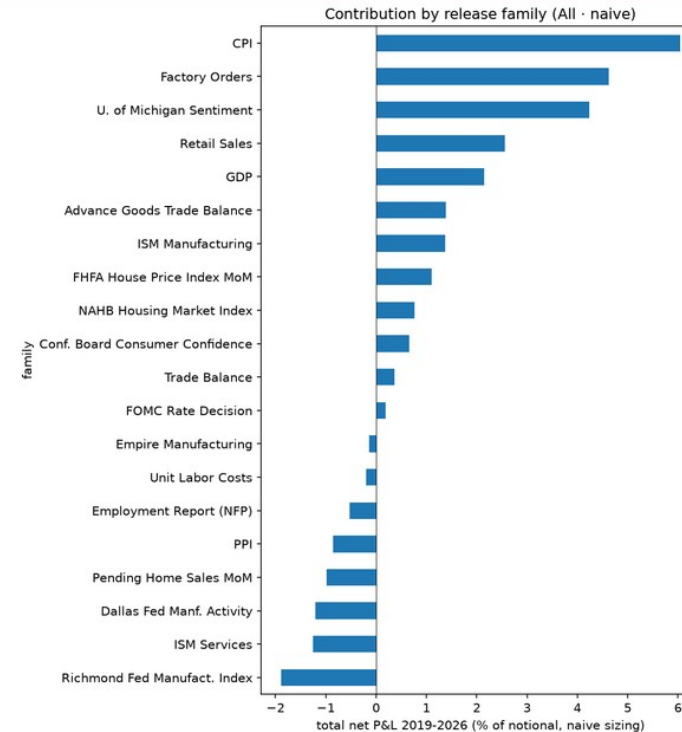
- ES and NQ: the naive all-release book climbs steadily, biggest gains in 2022 but positive outside it (NQ +17.5%, ex-2022 Sharpe 0.44; ES +12.8%, 0.28)
- Volatility sizing adds P&L but raises drawdowns to 12–14%
- ZN: every book falls almost linearly, paying the cost on each trade

# RQ5 · CPI leads; the rest is diversification across small edges

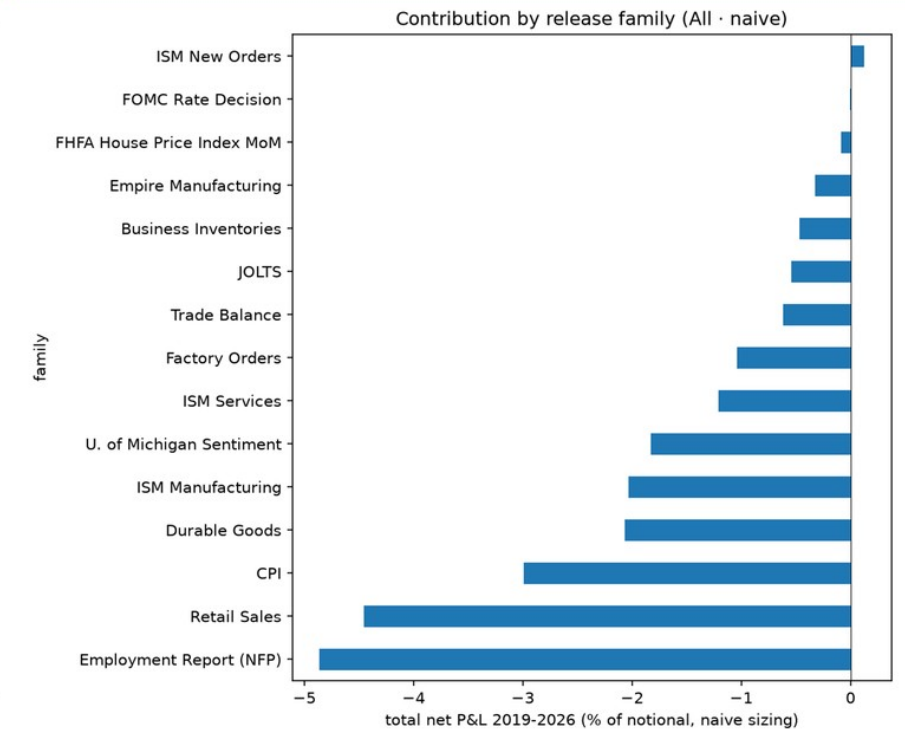
ES · E-mini S&P 500



NQ · E-mini Nasdaq-100



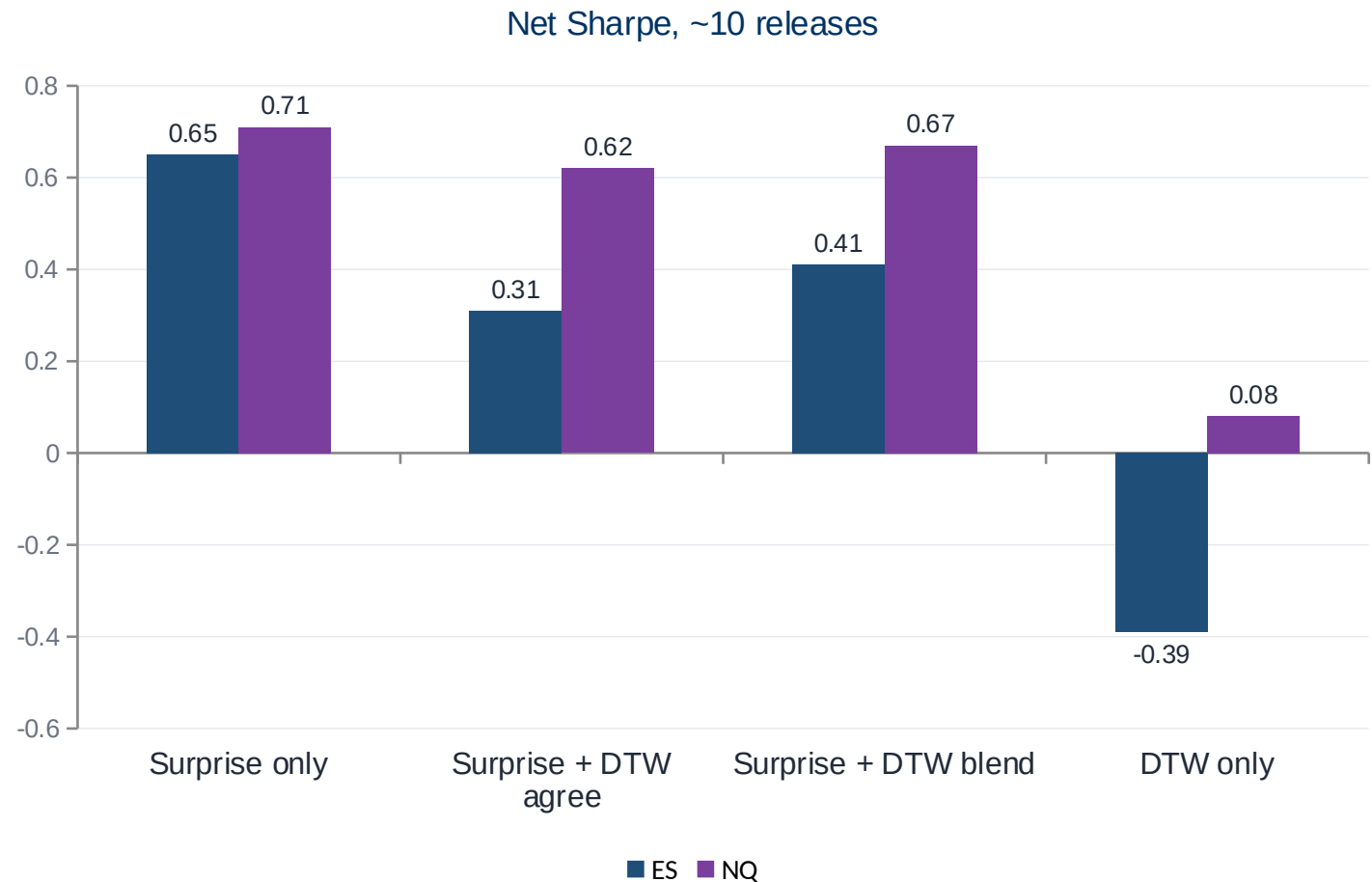
ZN · 10-year T-note



ES: CPI +4.3%, JOLTS +3.0%, Retail Sales +2.3% · NQ: CPI +6.0%, Factory Orders +4.6%, U. Michigan +4.2% · Both lose on the Employment Report and ISM Services · ZN: every major family loses about the cost on each trade

# RQ6 · DTW adds no directional information

Notebook 06: the team's DTW specification on the same releases, entry and costs



0.01 · 0.03 · 0.04

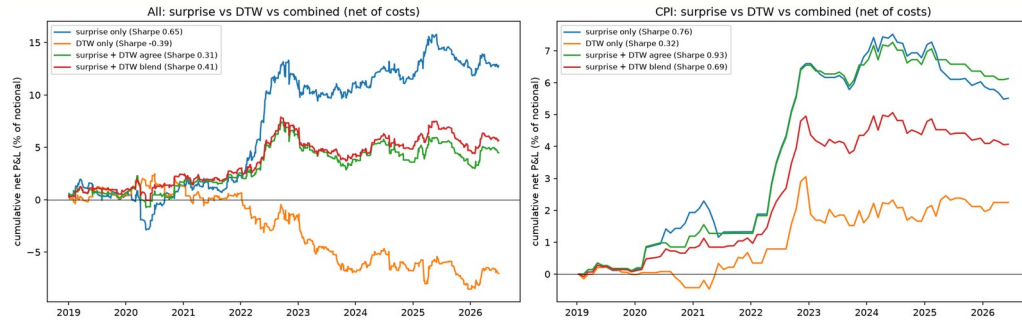
DTW rank IC with the 30-min return (ES · NQ · ZN); useful signals show 0.05–0.10

## Takeaways

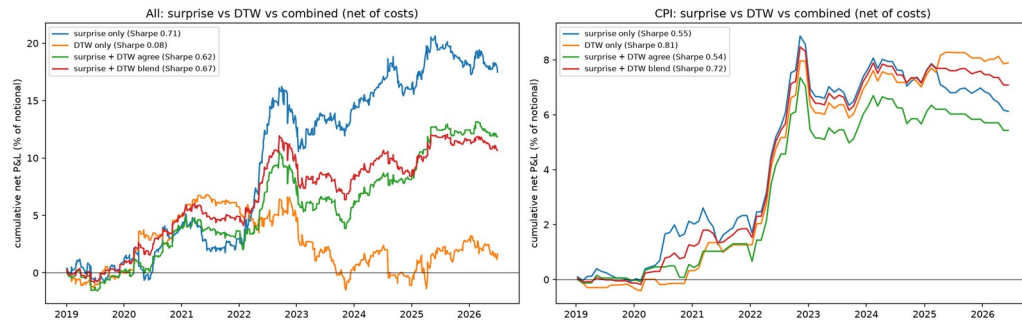
- Every combination lowers the surprise book's Sharpe
- CPI-only results point in different directions in ES and NQ: noise
- DTW is independent of the surprise (corr 0.02–0.11) but not complementary

# RQ6 · Surprise vs DTW over time

ES | E-mini S&P 500



NQ | E-mini Nasdaq-100



ZN | 10-year T-note

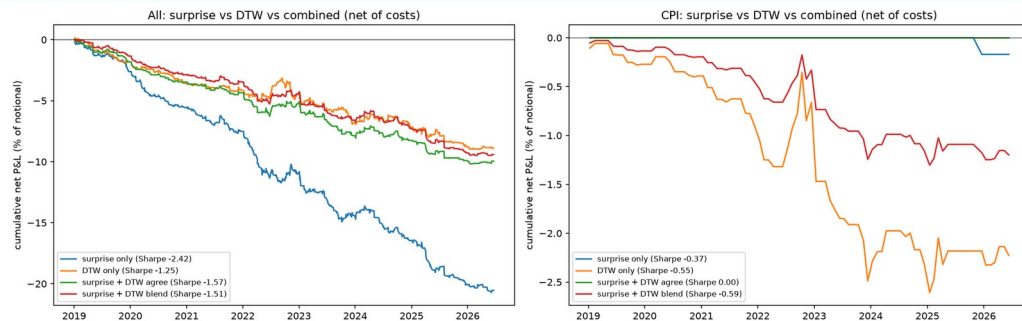
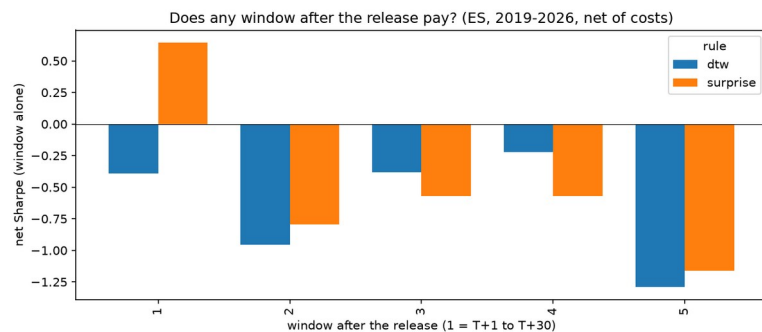


Figure 27 · ~10 releases (left panels) and CPI (right panels)

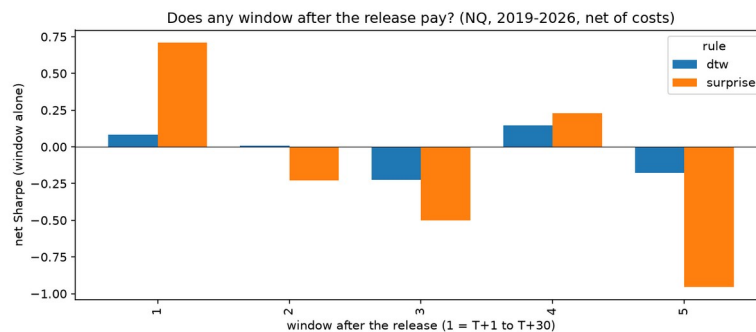
- ES: surprise only rises in steps (mostly 2022); DTW only is flat until 2022, then falls to about -7%
- NQ: DTW only rises in 2019–2022, then gives it all back from 2023, a pattern that does not persist
- The surprise-only line is the highest in every equity panel for ~10 releases
- CPI: the four lines move together; the differences are small and point in different directions in ES and NQ
- ZN: every line falls almost linearly (the cost); the surprise line falls fastest because it trades most

# RQ6 · More windows add trades, not profit

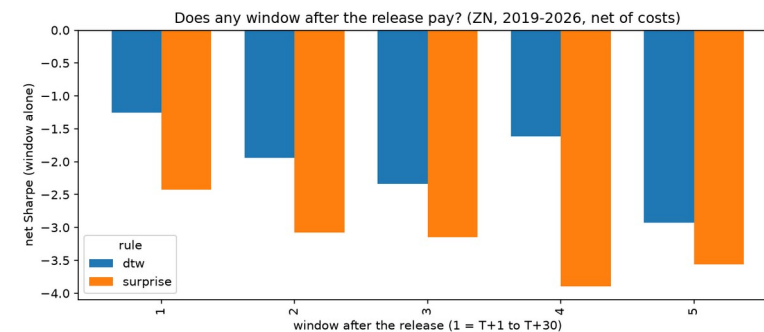
ES | E-mini S&P 500



NQ | E-mini Nasdaq-100



ZN | 10-year T-note



| Trades/yr, Sharpe           | ES         | NQ         |
|-----------------------------|------------|------------|
| Surprise, window 1 only     | 98, +0.65  | 97, +0.71  |
| Surprise W1 + DTW W2-5      | 348, -0.75 | 342, +0.27 |
| Surprise ladder (5 windows) | 490, -1.00 | 484, -0.29 |
| DTW ladder (5 windows)      | 314, -1.23 | 309, -0.07 |
| Peer DTW slide (ES)         | 185, +0.61 | —          |

## Five windows after each release (peer structure)

- Surprise rank IC: 0.14 in window 1 (ES),  $\approx 0$  in windows 2-5
- The ceiling is  $\sim 100$  trades a year per market at one trade per release
- The right way to trade more is more markets with a drift, not more windows

# RQ6 · Ladder books: turnover and cost, not profit

ES | E-mini S&P 500



**+12.8%**

ES window 1 only: the only ES book that ends positive

**-29% to -48%**

ES ladder books over 2019-2026

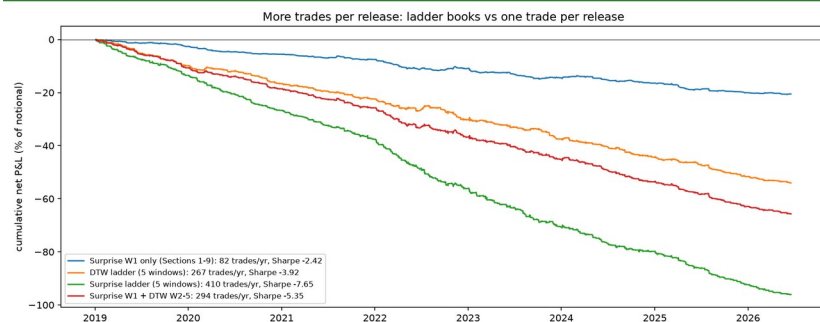
NQ | E-mini Nasdaq-100



## Figure 29 · three ladder books vs window 1 only

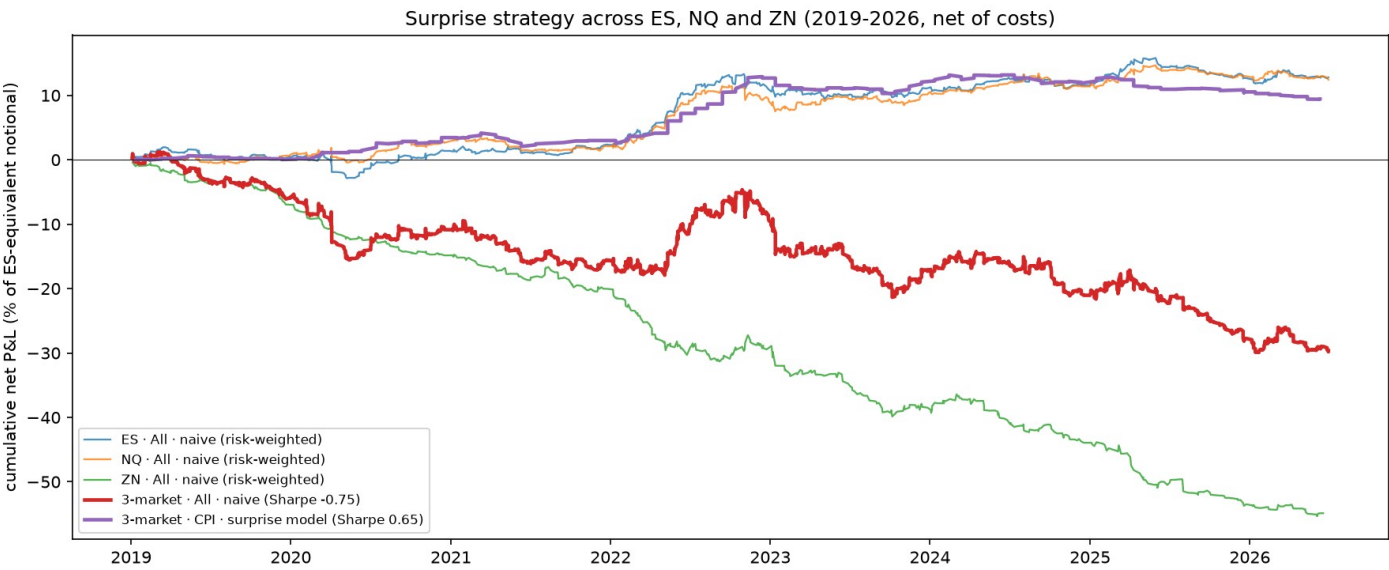
- ES: each extra window adds about 1-2 bps of cost with no edge
- NQ: cost below 1 bp, so 'surprise W1 + DTW W2-5' ends +12.9%, still below window 1 alone (+17.5%) with a drawdown three times larger
- ZN: the ladder books lose 54% to 96% of notional
- More windows per release add turnover and cost, not profit

ZN | 10-year T-note



# RQ7 · ES + NQ combine well; ZN destroys the portfolio

Notebook 07: equal-risk weights from history only



195

trades a year, ES + NQ

0.72

net Sharpe (1.01 at ¼ tick)

+25%

total net P&L, 2019–26

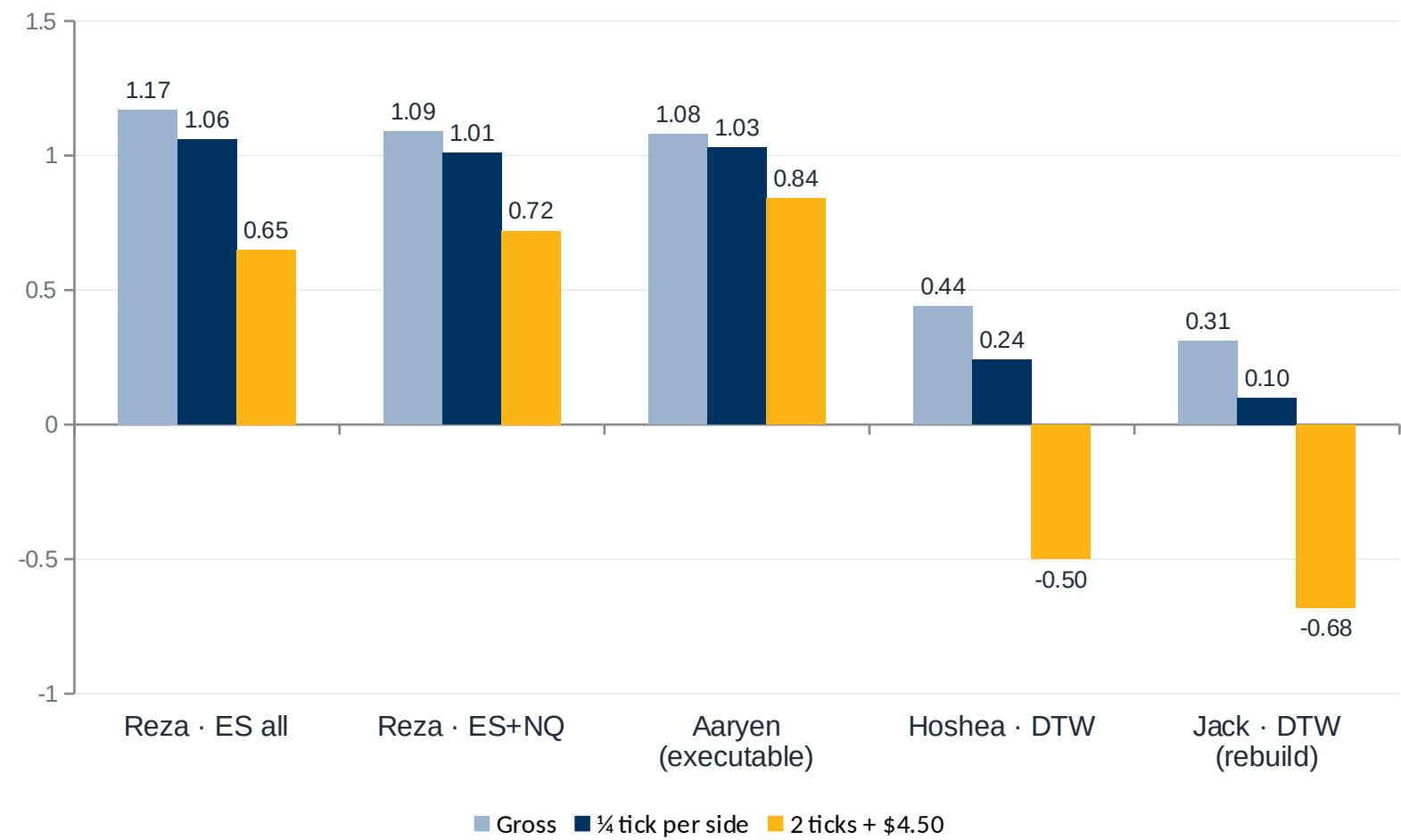
-0.75

Sharpe with ZN added

Signs are economically consistent across markets, but ZN has no drift and gets the largest risk weight.

# RQ7 · The team's four workstreams at equal costs

Notebook 08: each person's own trades re-priced on the same window (2019–2026) and Sharpe method



## Takeaways

- 300+ trade-a-year DTW books are positive only at 1/4 tick
- Surprise-based books survive realistic costs
- Reza: highest total return at every cost level
- Jack's rebuild (310 tr/yr, 0.16) is an approximation, not his slide

# RQ7 · All four workstreams, three cost levels

Figure 33 · Gross (no costs)

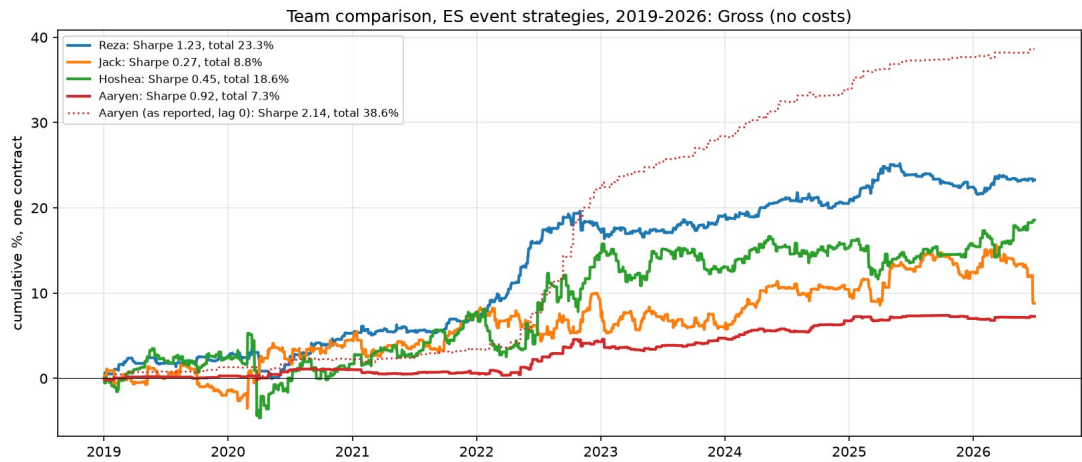


Figure 34 · Jack's cost: ¼ tick per side

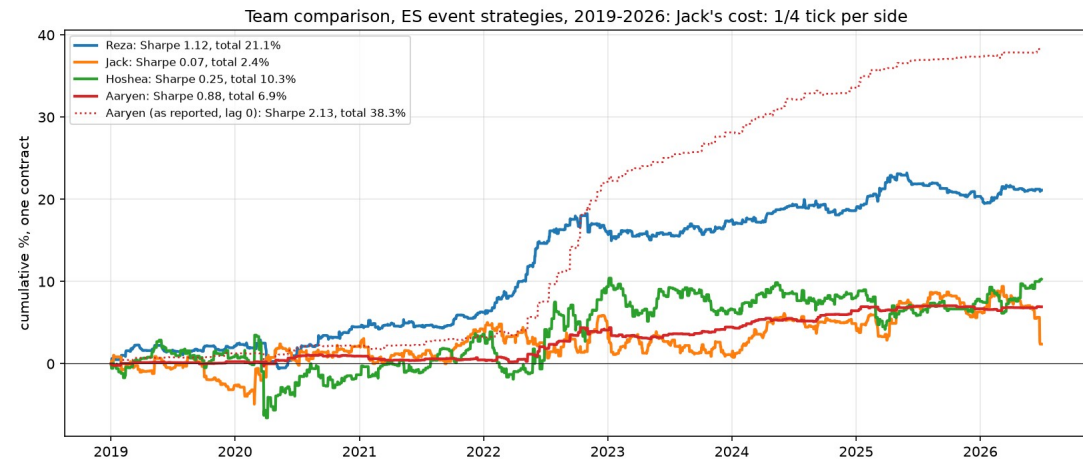
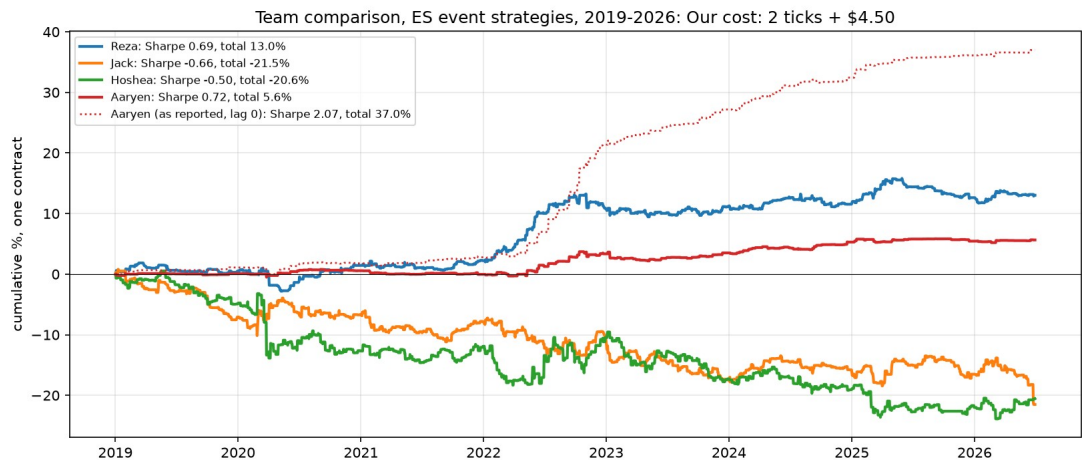


Figure 35 · Our cost: 2 ticks + \$4.50



| Sharpe / total       | Gross        | ¼ tick       | 2 ticks+\$4.50 |
|----------------------|--------------|--------------|----------------|
| <b>Reza (ES all)</b> | 1.23 / 23.3% | 1.12 / 21.1% | 0.69 / 13.0%   |
| Jack (rebuild)       | 0.27 / 8.8%  | 0.07 / 2.4%  | -0.66 / -21.5% |
| Hoshea               | 0.45 / 18.6% | 0.25 / 10.3% | -0.50 / -20.6% |
| Aaryen (exec.)       | 0.92 / 7.3%  | 0.88 / 6.9%  | 0.72 / 5.6%    |
| Aaryen (reported)    | 2.14 / 38.6% | 2.13 / 38.3% | 2.07 / 37.0%   |

Only the surprise books stay positive at our cost; the DTW books fall to about -20%.

# RQ7 · Each workstream in the peer slide format (1/2)

Figure 36 · Reza, ES all releases

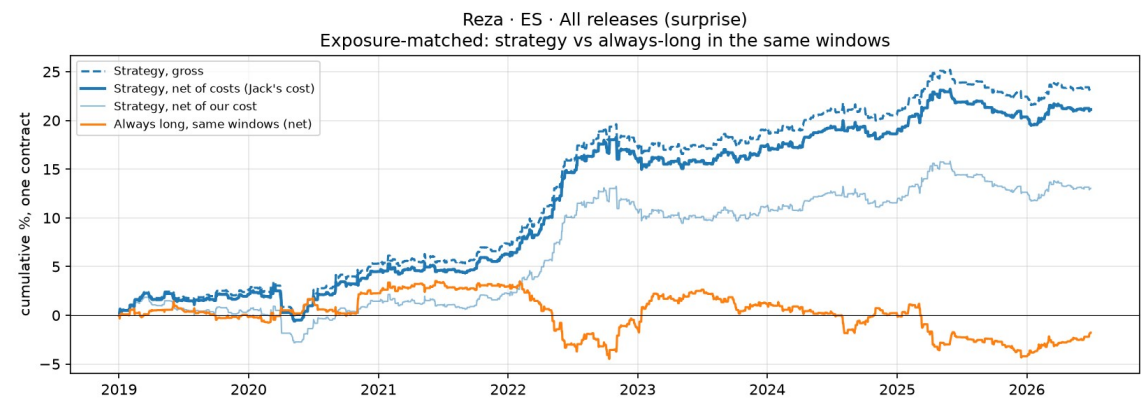


Figure 37 · Reza, ES + NQ all releases

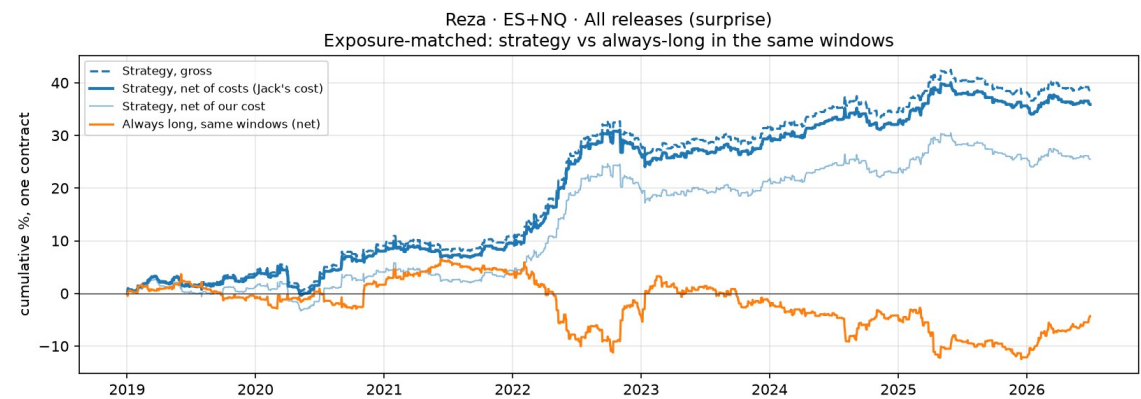
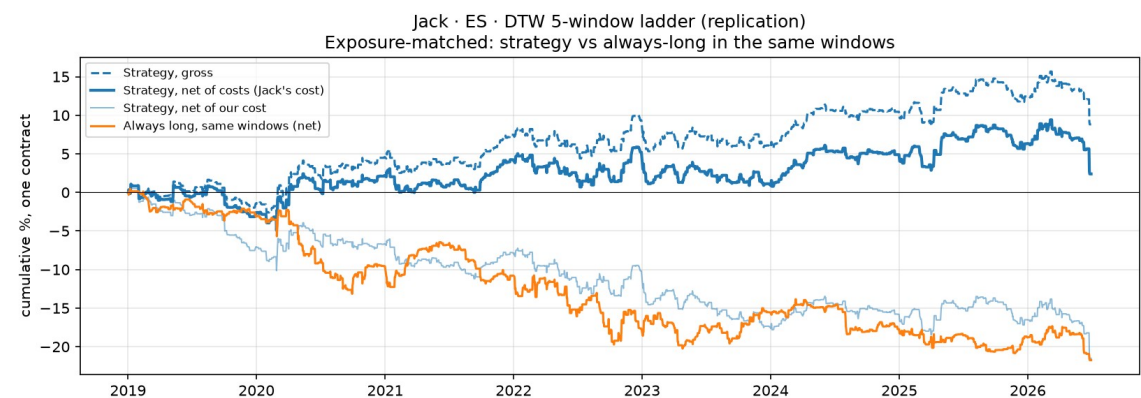


Figure 38 · Jack, DTW ladder (rebuild)



| ¼ tick per side           | Reza ES | Reza ES+NQ | Jack (rebuild) |
|---------------------------|---------|------------|----------------|
| Total %                   | 21.1    | 35.9       | 2.4            |
| Daily Sharpe              | 1.12    | 1.06       | 0.07           |
| Max drawdown              | -3.7%   | -7.0%      | -7.0%          |
| Trades / yr               | 98      | 195        | 315            |
| Always long, same windows | -1.8%   | -4.3%      | -21.7%         |

Jack's own slide (his data, 2018–26): 185 trades/yr, Sharpe 0.61, 18.6%. Our rebuild is an approximation.

# RQ7 · Each workstream in the peer slide format (2/2)

Figure 39 · Hoshea, DTW event games

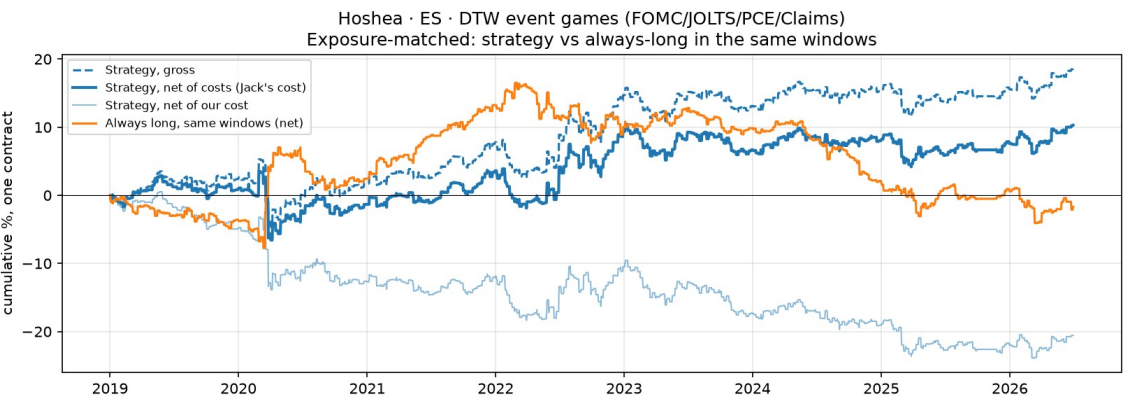


Figure 40 · Aaryen, classifier, as reported (entry at pre-release price)

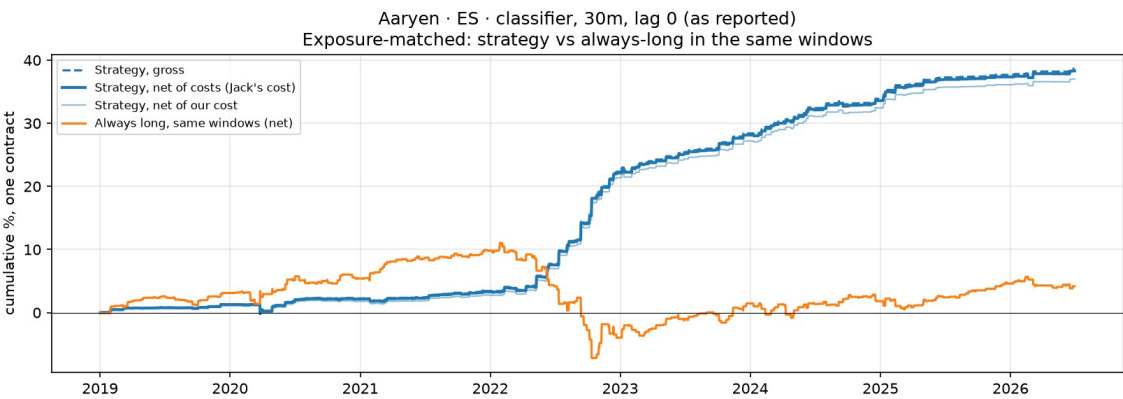
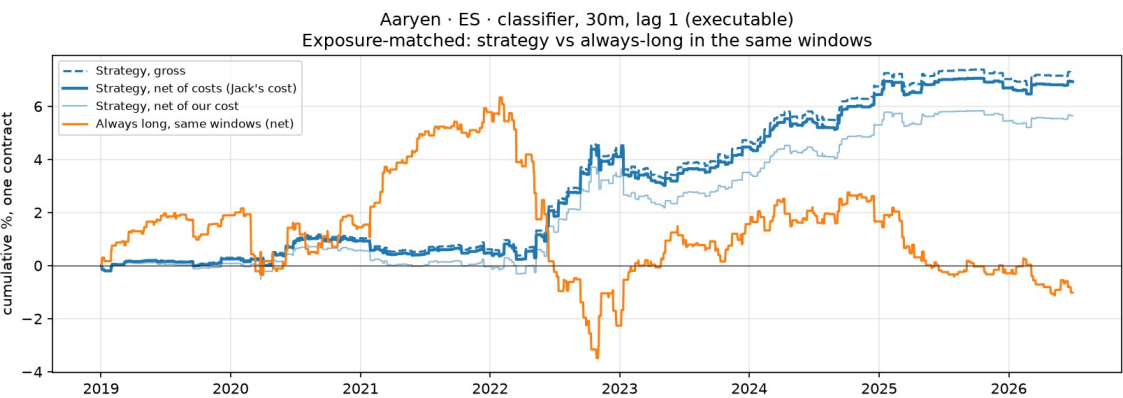


Figure 41 · Aaryen, classifier, executable (entry 1 min later)

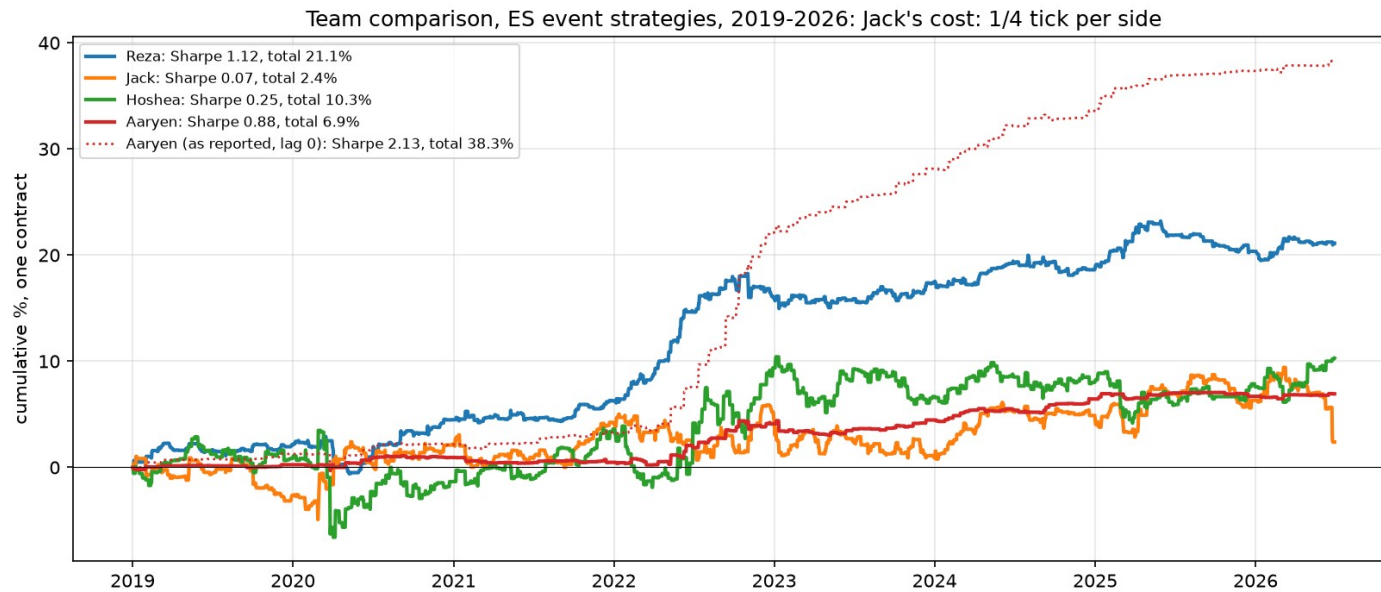


| ¼ tick per side           | Hoshea | Aaryen reported | Aaryen exec. |
|---------------------------|--------|-----------------|--------------|
| Total %                   | 10.3   | 38.3            | 6.9          |
| Daily Sharpe              | 0.25   | 2.13            | 0.88         |
| Max drawdown              | -10.1% | -1.5%           | -1.4%        |
| Trades / yr               | 372    | 55              | 55           |
| Always long, same windows | -1.7%  | +4.2%           | -1.0%        |

Same positions, entry one minute apart: 38.3% → 6.9%. ~80% of the reported profit is the first-minute jump.

# The decisive assumption: when you enter

Aaryen's 30-minute classifier, identical positions, entry moved by one minute



**2.1 → 0.9**

Sharpe: entry at the pre-release price vs one minute after the release

**38.6% → 7.3%**

Gross total return: ~80% of the reported profit is the first-minute jump

Report every strategy with an executable entry and at two cost levels. These choices change Sharpe ratios more than any model choice.

# Discussion: why equities and not bonds?

1

## Interpretation takes time

A CPI print maps into yields almost mechanically; for equities investors must also judge growth, margins and risk premia. NQ (long duration) drifts most.

2

## Liquidity returns gradually

Order books thin out at the release and refill over minutes: part of the drift is price discovery.

3

## Bonds move first

Treasury futures price rate news directly and fastest; by T+1 the move is complete.

4

## Cost relative to move

ZN: 3 bps cost vs 10 bps average CPI drift, 0.8 bps predictable. ES/NQ: 1.7 / 0.7 bps vs 20–26 bps.

# Final comparison of the three model families

| Model family             | Best result (net, 2019–2026)                                                          | Verdict                                        |
|--------------------------|---------------------------------------------------------------------------------------|------------------------------------------------|
| Surprise only            | CPI $\approx 1.0$ (ES); 10 releases 0.65 (ES), 0.71 (NQ); ES+NQ 0.72 at 195 trades/yr | <b>Recommended</b>                             |
| Surprise + State         | CPI 0.77 (ES), 0.75 (NQ)                                                              | Worse out of sample; use state for risk sizing |
| Surprise + Pattern (DTW) | 0.31–0.41 (ES), 0.62–0.67 (NQ)                                                        | Worse; no directional information              |
| Five-window ladder       | –0.75 to –1.23 (ES)                                                                   | Adds only losing trades                        |
| Any ZN strategy          | –0.3 to –2.4                                                                          | Not tradeable at realistic costs               |

*Only the macro surprise in the first 30 minutes after a release, in equity-index futures, gives a robust, tradeable directional edge.*

# Recommendations

## 1 Trade rule

Enter at the close of the first post-release minute, exit at T+30; sign of the family surprise; no DTW, no extra windows

## 2 Two-leg book

CPI core leg (surprise model or signal  $\times$  vol) + multi-release naive leg on ES and NQ (~195 trades/yr)

## 3 Risk, not direction

Scale position size by expected move (20-day vol, drawdown); stops in units of expected move

## 4 Clean universe

Drop families with no real surprise (FOMC, IORB); fix one CPI definition; skip  $|z| < 0.5$

## 5 Regime scaling

Raise exposure when macro data drive policy expectations (e.g. CPI-2y yield correlation)

## 6 Execution & monitoring

Paper-trade 3-6 months; limit orders; rolling IC and Sharpe with pre-set stop rules

# Future work

## Fixed CPI definition

Pre-register one definition and a  $|z|$  threshold; test on 2026+ hold-out

## Order-book execution

Level-2 data to measure fills; entries at 10–60 seconds

## Live paper trading

Machine-readable release feed; measure latency and slippage

## More equity markets

RTY, YM, Euro Stoxx 50, Nikkei around US releases

## Short-rate markets

2-year Treasury (ZT), SOFR futures: is there a drift?

## Volatility forecasting

State and DTW dispersion for sizing, stops, options

## Richer surprises

Revisions, forecast dispersion, market-implied expectations

## Longer sample

Back to 2000 across monetary-policy regimes

# Thank you

**Benjamin Steel (BlackRock)** proposed the research question and supervised the project.

**Ian Kaufman and Chris Pohalski (UC Berkeley Haas, MFE program)** coordinated the partnership with BlackRock and the project meetings.

**Jack Duncan, Hoshea Zeng and Aaryen Mehta** for the DTW strategies, release screening and surprise classification credited throughout.

Code, notebooks, results and the full report: [github.com/jack-duncan/blackrock-intraday](https://github.com/jack-duncan/blackrock-intraday) (Reza/)

*Student industry project. Views are the authors' own and do not represent BlackRock or UC Berkeley. Not investment advice.*